

Data Visualization

MKT 566 · Fall 2026

Davide Proserpio

USC Marshall School of Business

Before We Start

Recap: your toolkit

Everything in this course runs on three pieces from Tuesday's setup guides:

1. **VS Code**: the editor where everything happens
2. **An AI assistant** inside it: [Claude Code](#), [Codex](#), [Copilot](#), or [Cursor](#)
3. **R + the R extension**: open a script, run it one line at a time with [Cmd+Enter](#) (Mac) / [Ctrl+Enter](#) (Windows); charts appear in a VS Code tab

Ready check: open a course script, press [Cmd+Enter](#) on a chart line, and a chart appears in a tab. Not there yet? Follow the [VS Code + AI setup guide](#) and [Running R Code in VS Code](#), or bring your laptop to office hours.

Vibecoding: how you will actually write code

- **Vibecoding** (Karpathy, 2025): describe what you want in plain English, let the AI write the code, run it, look at what comes out, refine, repeat
- Most code in this class (and, increasingly, in industry) gets written this way. That changes *which* skills are scarce:
 - **Asking precisely**: “a histogram of weekly sales, 30 bins” beats “make a chart”
 - **Reading results**: does the chart answer the question? Is the number plausible?
 - **Debugging by conversation**: paste the error, ask why, ask for the fix
- What it does not change: **you own everything you submit**, including the AI’s mistakes

Today builds the vocabulary this workflow needs: once you know the chart types by name and what each is for, you can ask for the right one **and judge what you get back**.

Reminder: every assignment includes a log of the LLM prompts you used.

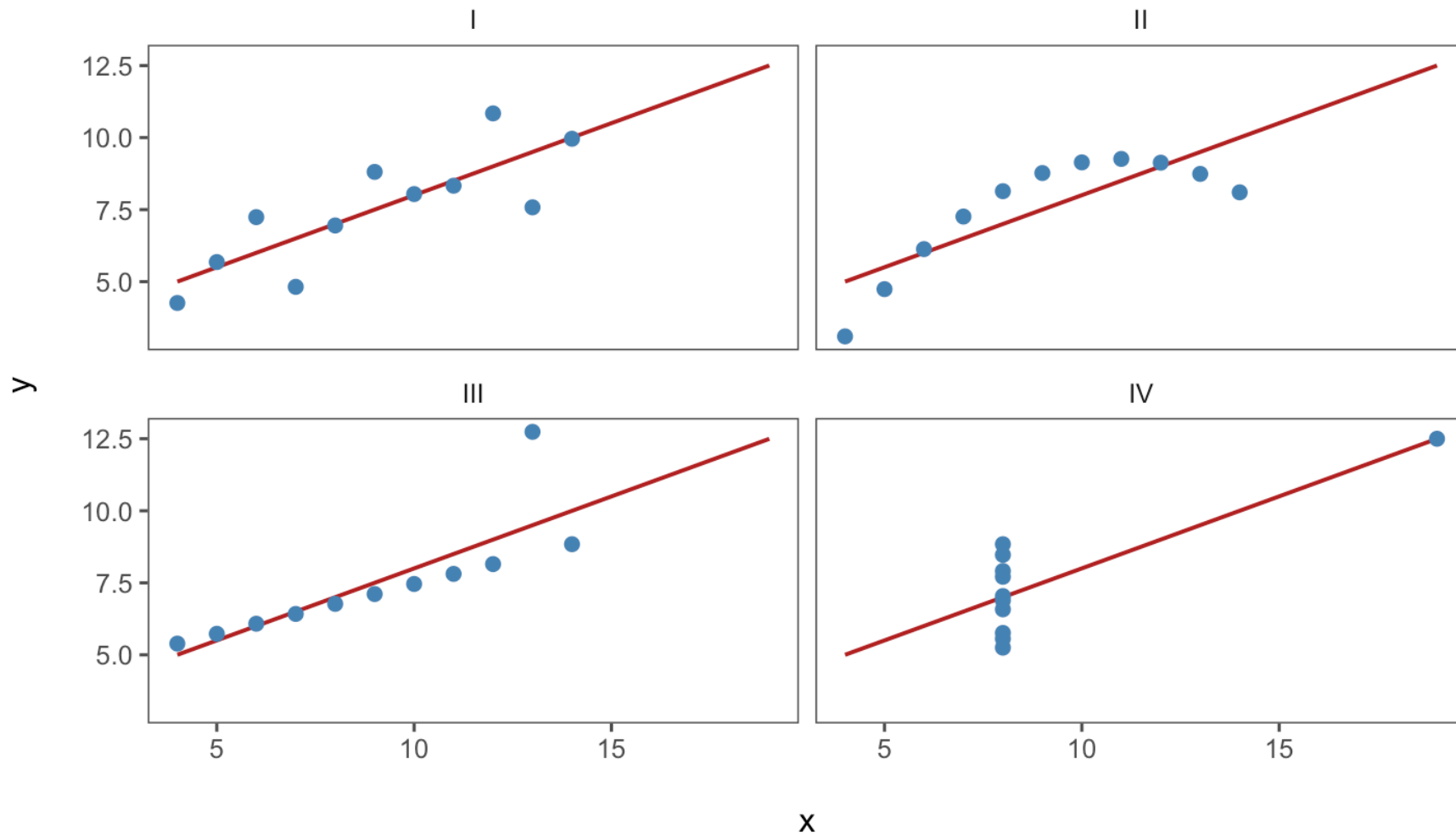
Why Visualize?

Why visualize?

“The simple graph has brought more information to the data analyst’s mind than any other device.” — John Tukey

- Summary statistics **hide** as much as they reveal
- A figure is usually the **fastest** way to communicate a result to a stakeholder
- In marketing analytics, most deliverables end life as a chart in someone’s deck

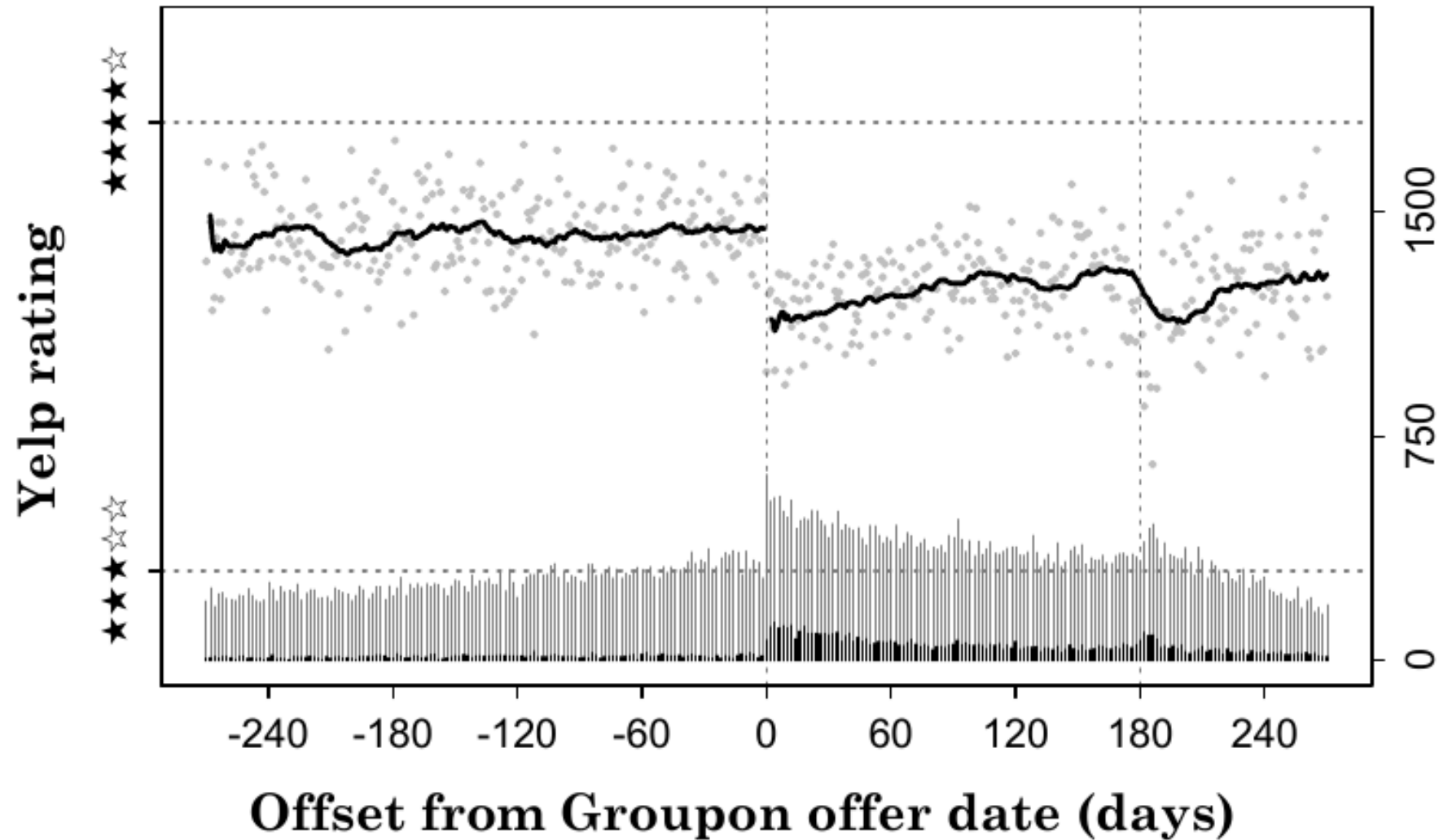
Same numbers, four different stories



Anscombe's quartet: all four datasets share the same means, variances, correlation ($r = 0.82$), and regression line ($y = 3 + 0.5x$).

Always plot your data first.

An (almost) perfect example



Source: [*The Groupon Effect on Yelp Ratings: A Root Cause Analysis*](#) (Byers, Mitzenmacher, and Zervas 2012), Figure 1a.

What we will learn

- What **types of charts** exist and what they are used for
- How to **pick the best visual** for different types of data
- How to create **compelling figures**

We will use **ggplot2**, an R library for data visualization ([intro to ggplot2](#)).

Content partially based on Chapter 3 of [R for Data Science](#).

R scripts for today

Two R scripts on the course website, in the [code/](#) folder:

1. [w1-2-chart-types-class.R](#): reproduces every chart type we discuss today
2. [w1-2-data-viz-beautify.R](#): a simple figure beautification process

Download the whole [code/](#) folder as a zip: [w1-code.zip](#) (it includes the datasets in [data/](#)), open the scripts in RStudio (or VS Code), and (try to) install the required libraries. These slides are themselves generated with Quarto + R: every chart you see is drawn live from those scripts' code.

Chart Types

A taxonomy

Family	Question it answers	Charts
Category comparison & composition	How do groups stack up? What makes up the total?	Bar, Pareto, treemap, pie, waterfall
Trends over time	How do values evolve or accumulate?	Line, area, bar + line combo
Distribution & density	What is the shape, spread, outliers?	Histogram, density, box, violin
Relationships	How do variables move together?	Scatter, bubble
Geospatial & matrix	How do values vary over space or grids?	Choropleth map, heatmap

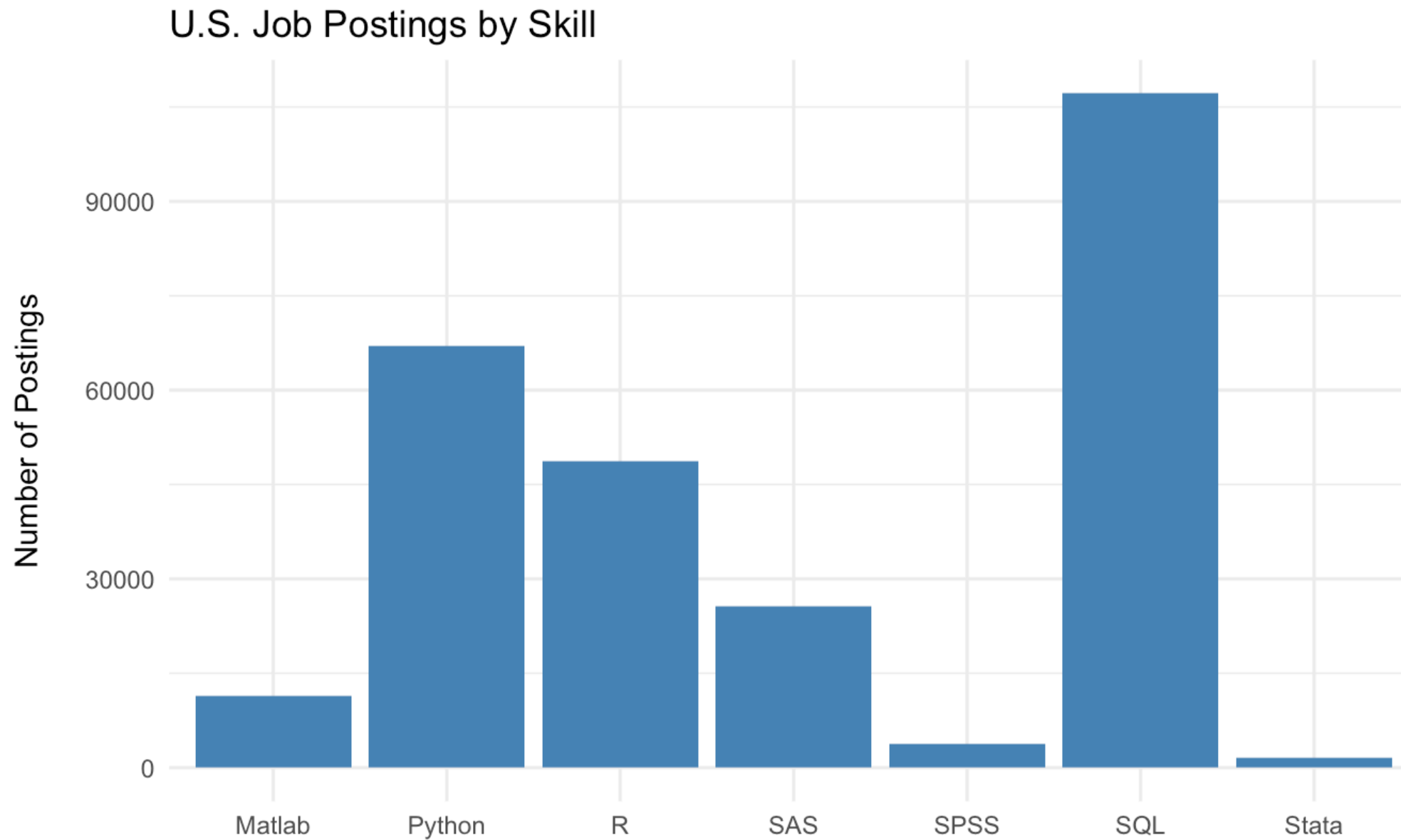
Category comparisons

Show how discrete groups or items stack up against one another.

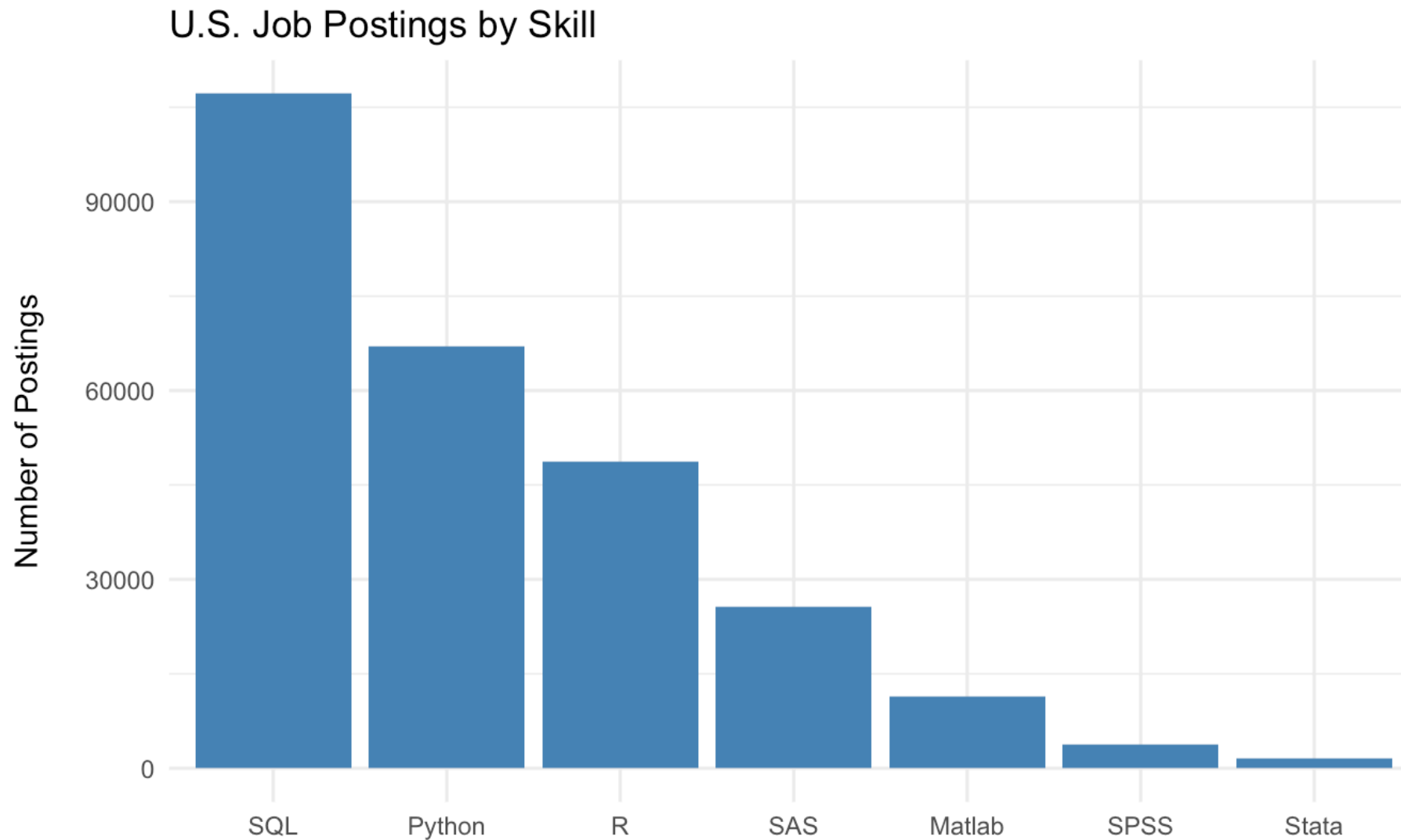
Running example: number of U.S. job postings mentioning each coding skill (`data/skills-postings.csv`).

- Bar chart
- Pareto chart (sorted bars + cumulative line)
- Treemap
- Pie chart
- Waterfall chart

Bar chart

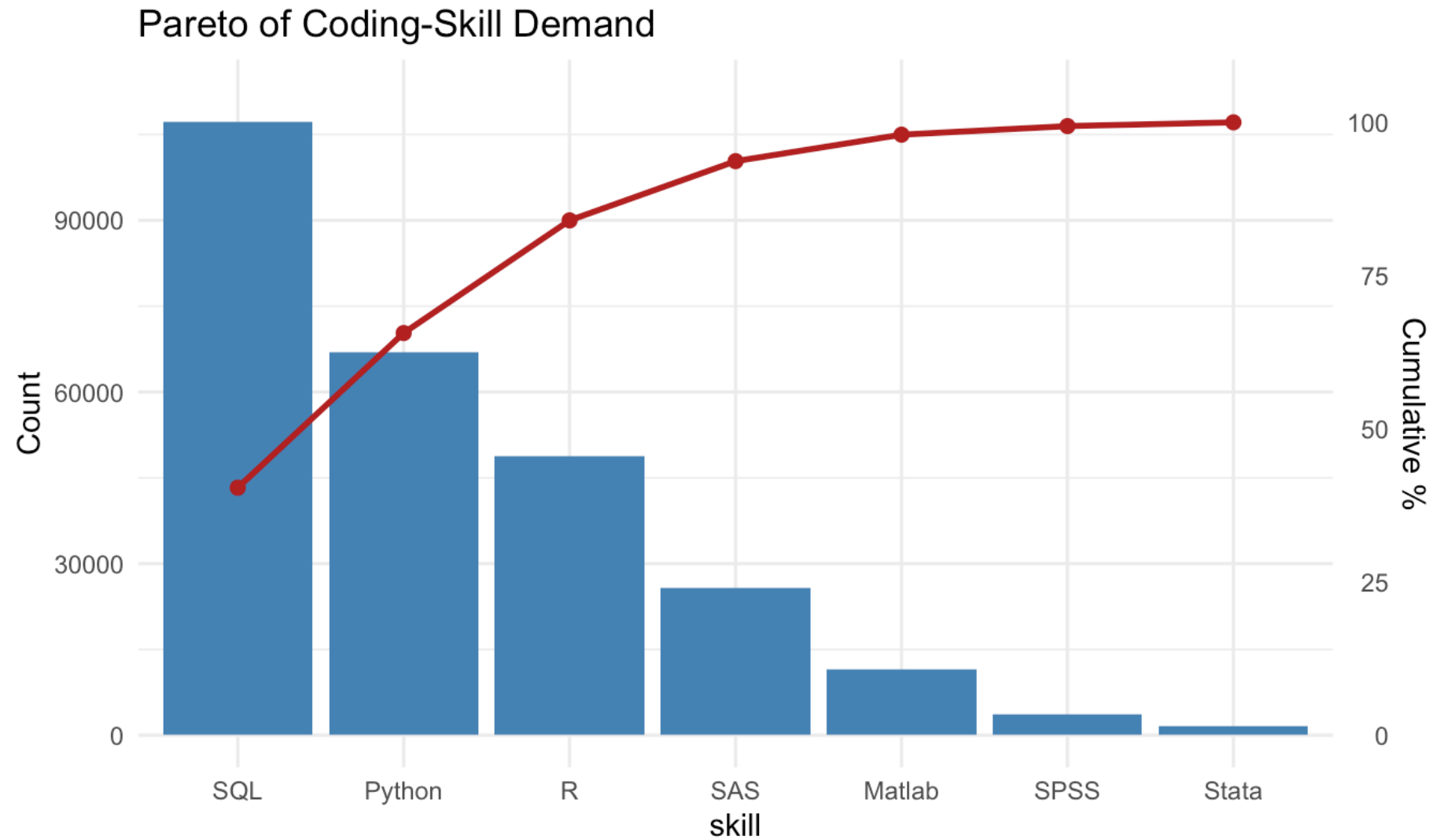


Bar chart: order your bars



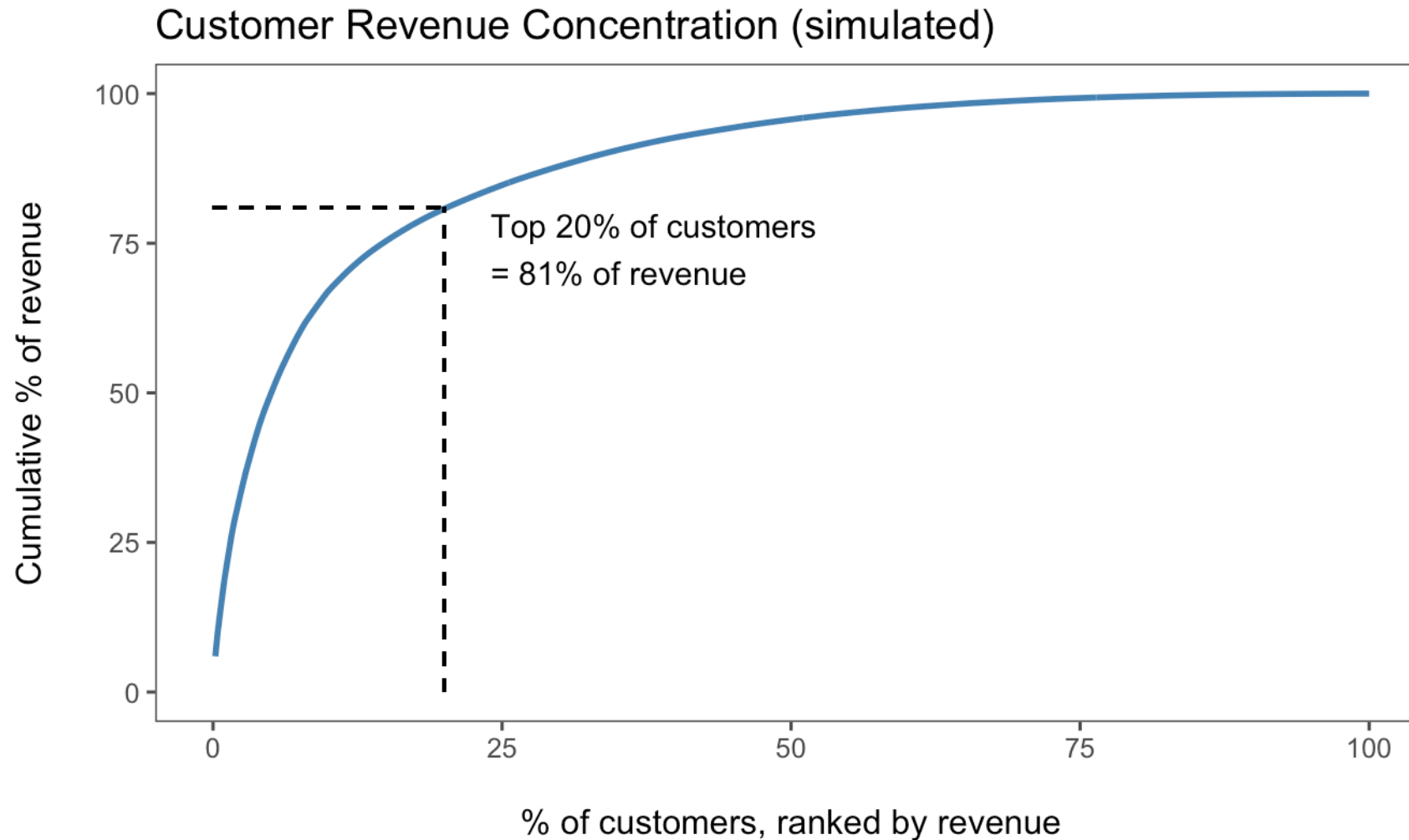
Alphabetical order is an accident; sorted order is a statement. `fct_reorder()` in R.

Pareto chart



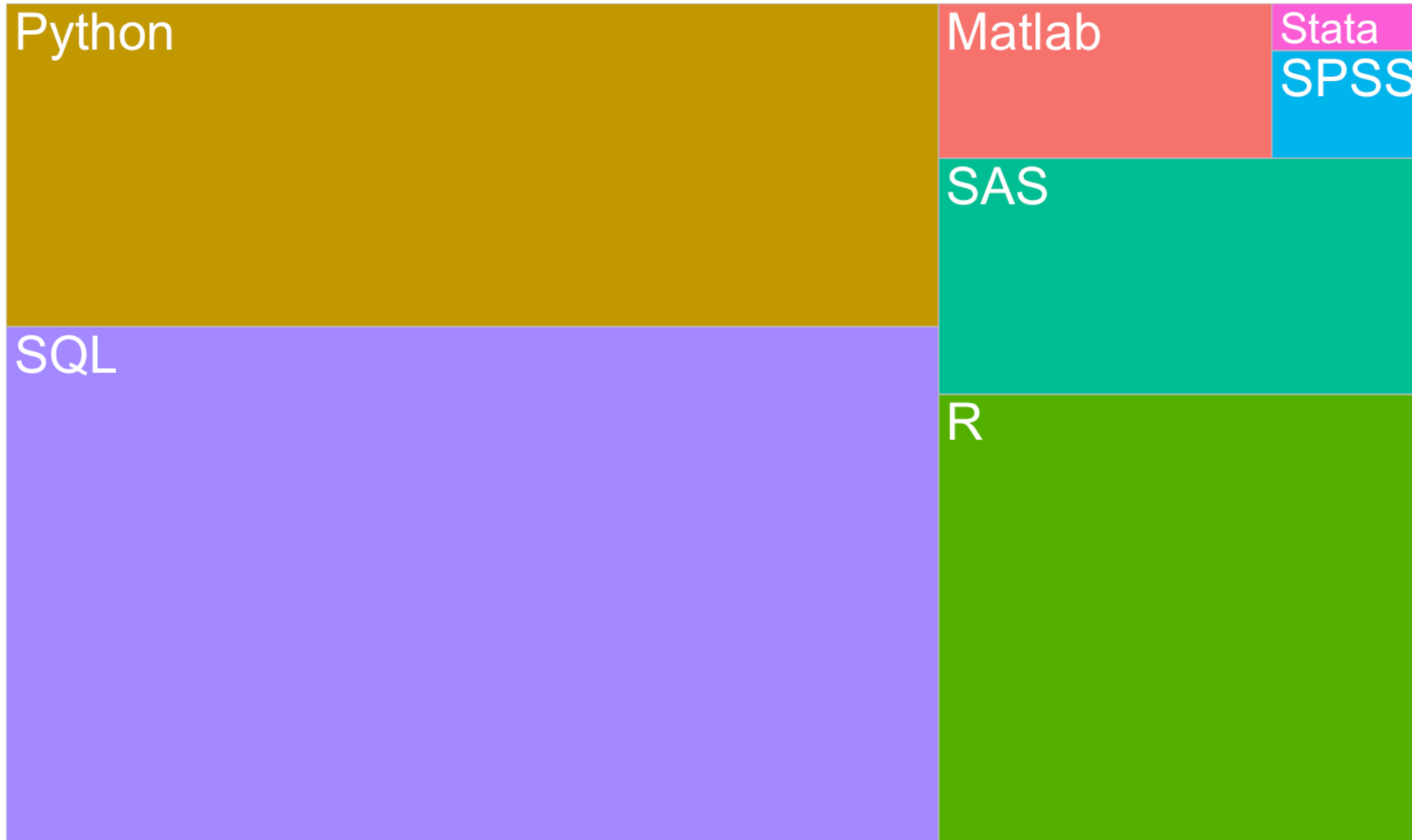
Sorted bars + cumulative share: identifies the “vital few” driving most of the total.

The 80/20 rule in marketing



Revenue is almost always concentrated in a small share of customers. This motivates segmentation, loyalty programs, and CLV analysis (all later in this course).

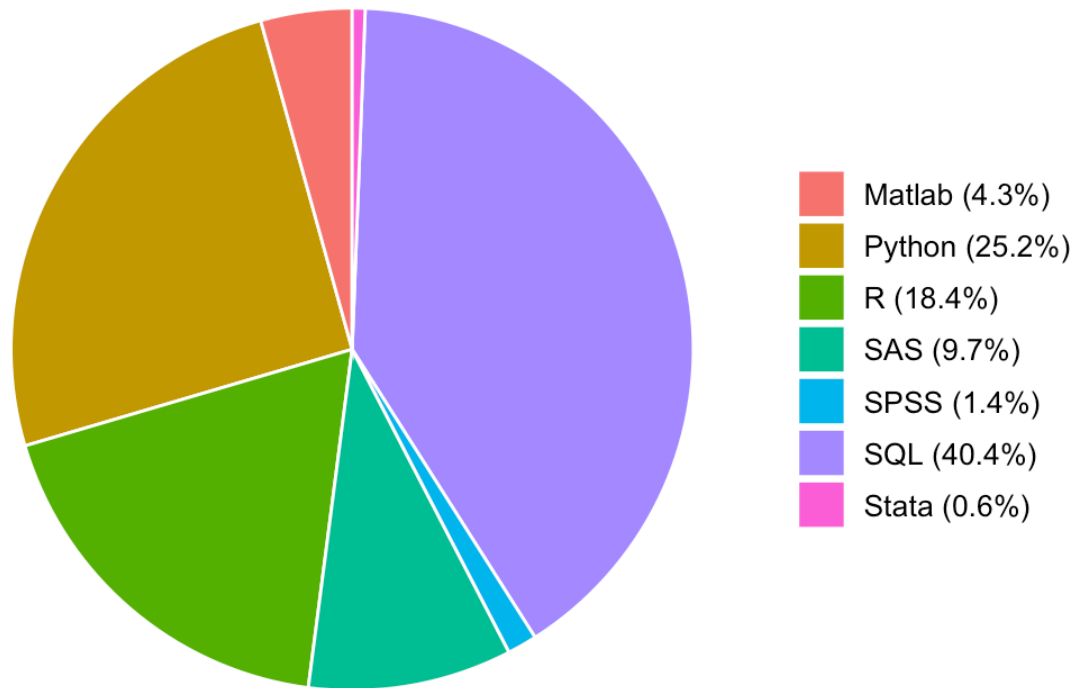
Treemap



Space-filling overview: area $\propto$ count. Good for hierarchies (category $\rightarrow$ brand $\rightarrow$ SKU).

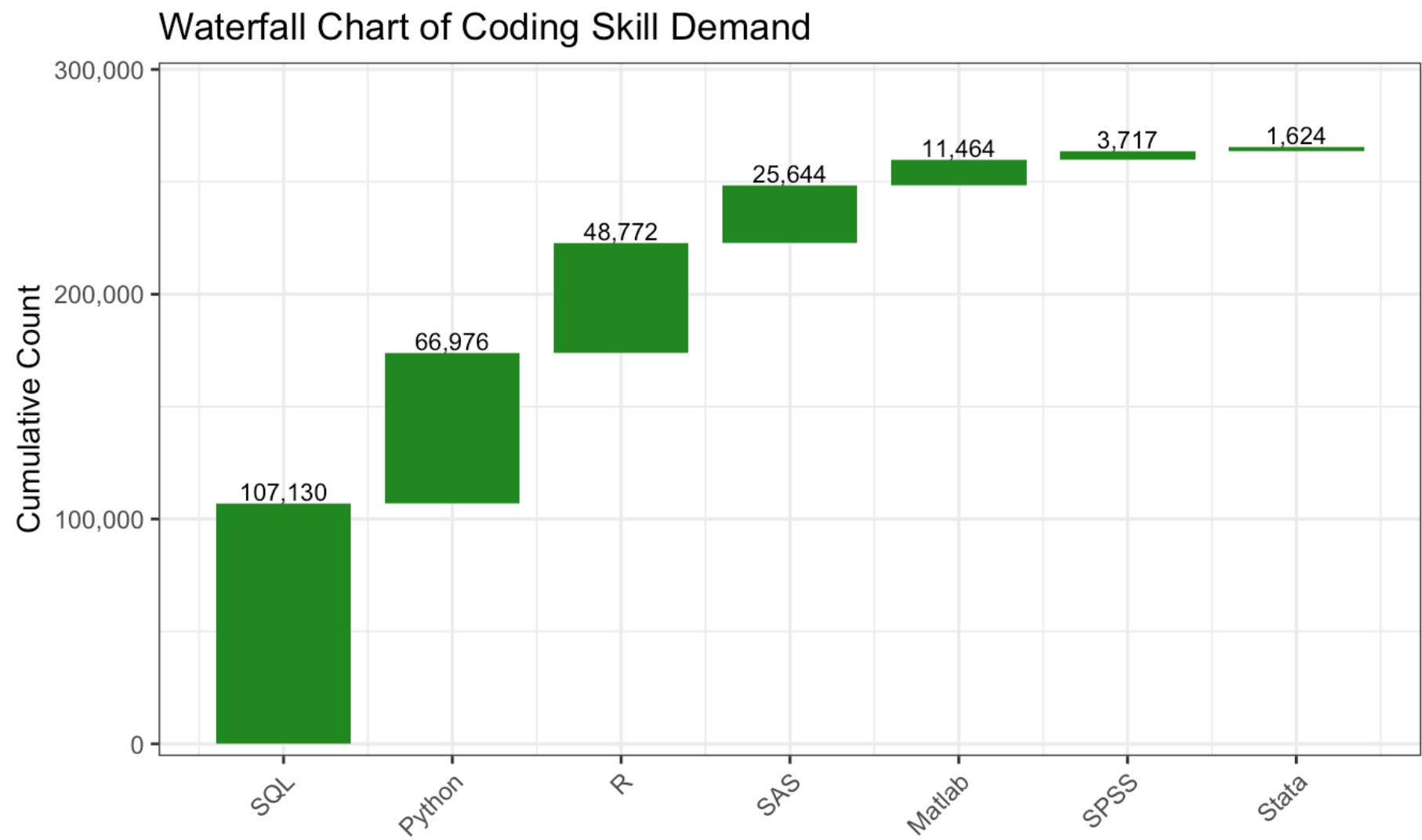
Pie chart

Market Share of Coding Skills



Use sparingly: humans compare **lengths** far better than **angles**. With more than 3–4 slices, a sorted bar chart is almost always clearer.

Waterfall chart



Each bar starts where the previous ended: shows incremental contributions to a total (common in revenue-bridge and budget decks).

Trends over time

Reveal how values evolve or accumulate.

- Line chart
- Area chart (stacked or cumulative)
- Bar + line combo

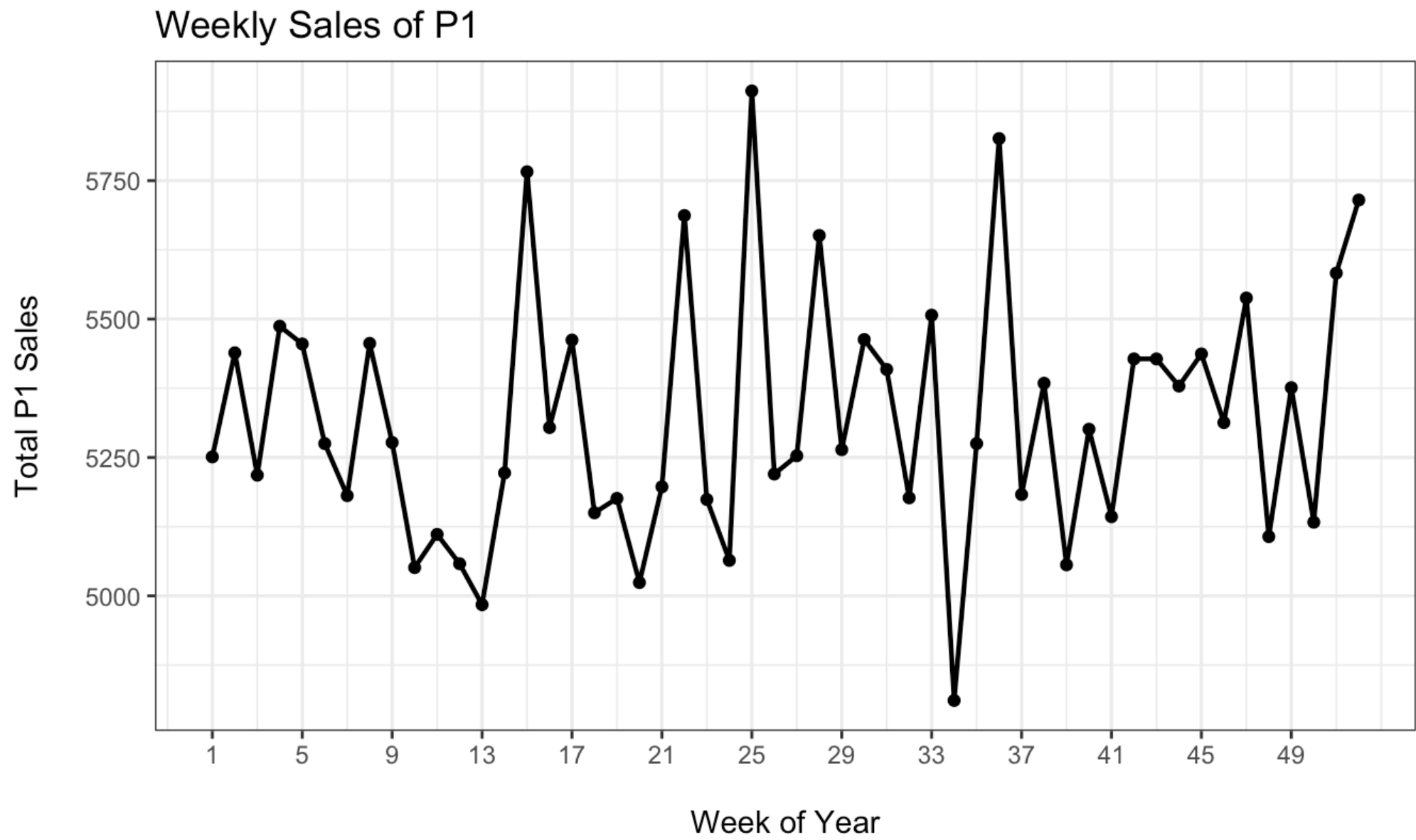
New dataset: store sales

```
1 store.df <- read.csv("data/store-sales.csv")
2 head(store.df)
```

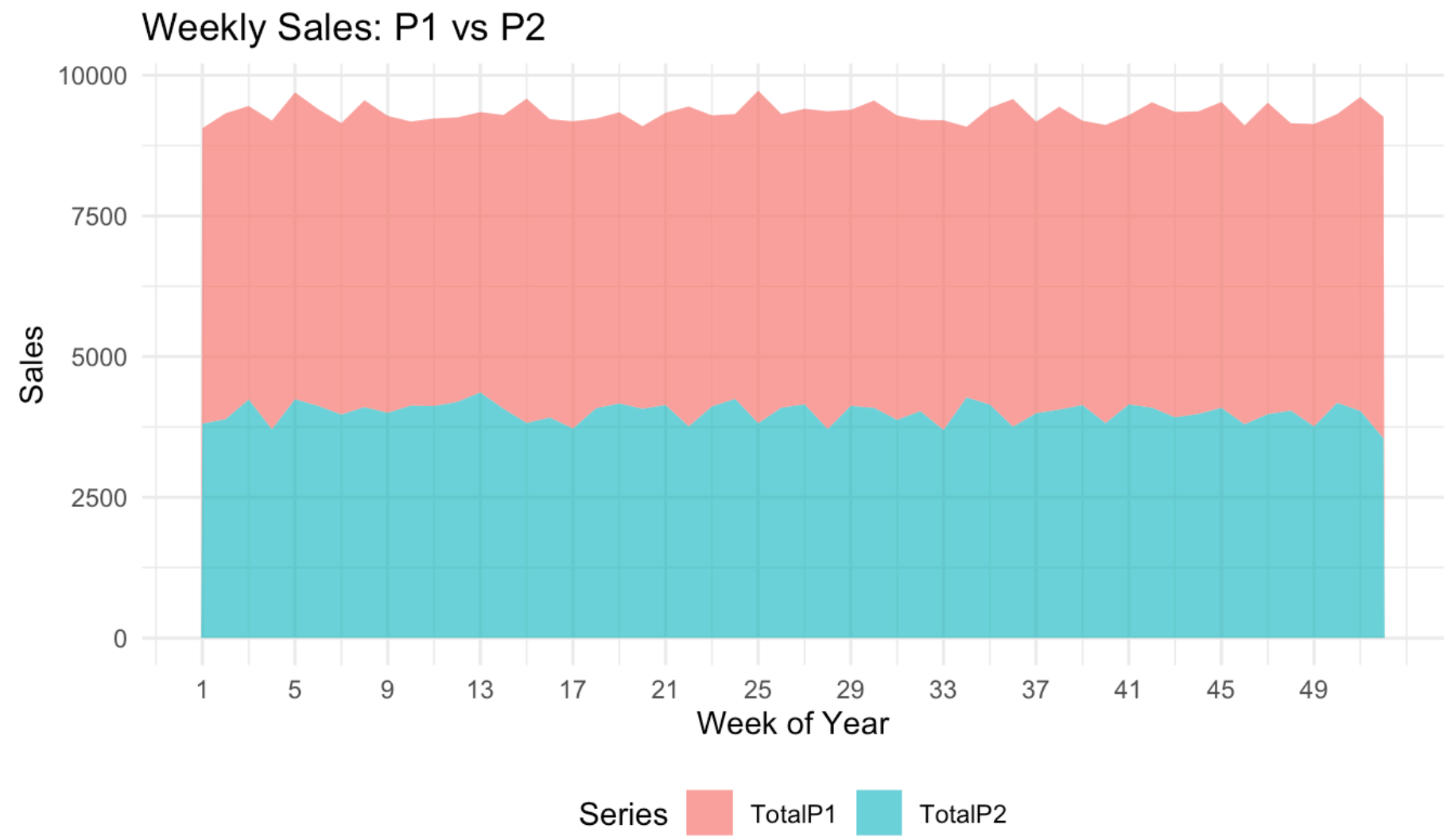
	storeNum	Year	Week	p1sales	p2sales	p1price	p2price	p1prom	p2prom	country
1	101	1	1	127	106	2.29	2.29	0	0	US
2	101	1	2	137	105	2.49	2.49	0	0	US
3	101	1	3	156	97	2.99	2.99	1	0	US
4	101	1	4	117	106	2.99	3.19	0	0	US
5	101	1	5	138	100	2.49	2.59	0	1	US
6	101	1	6	115	127	2.79	2.49	0	0	US

Two years of weekly sales of two products (P1, P2) across 20 stores in 7 countries, with prices and promotion flags. Sales are in **unit counts**. From Chapman & Feit, *R for Marketing Research and Analytics*.

Line chart

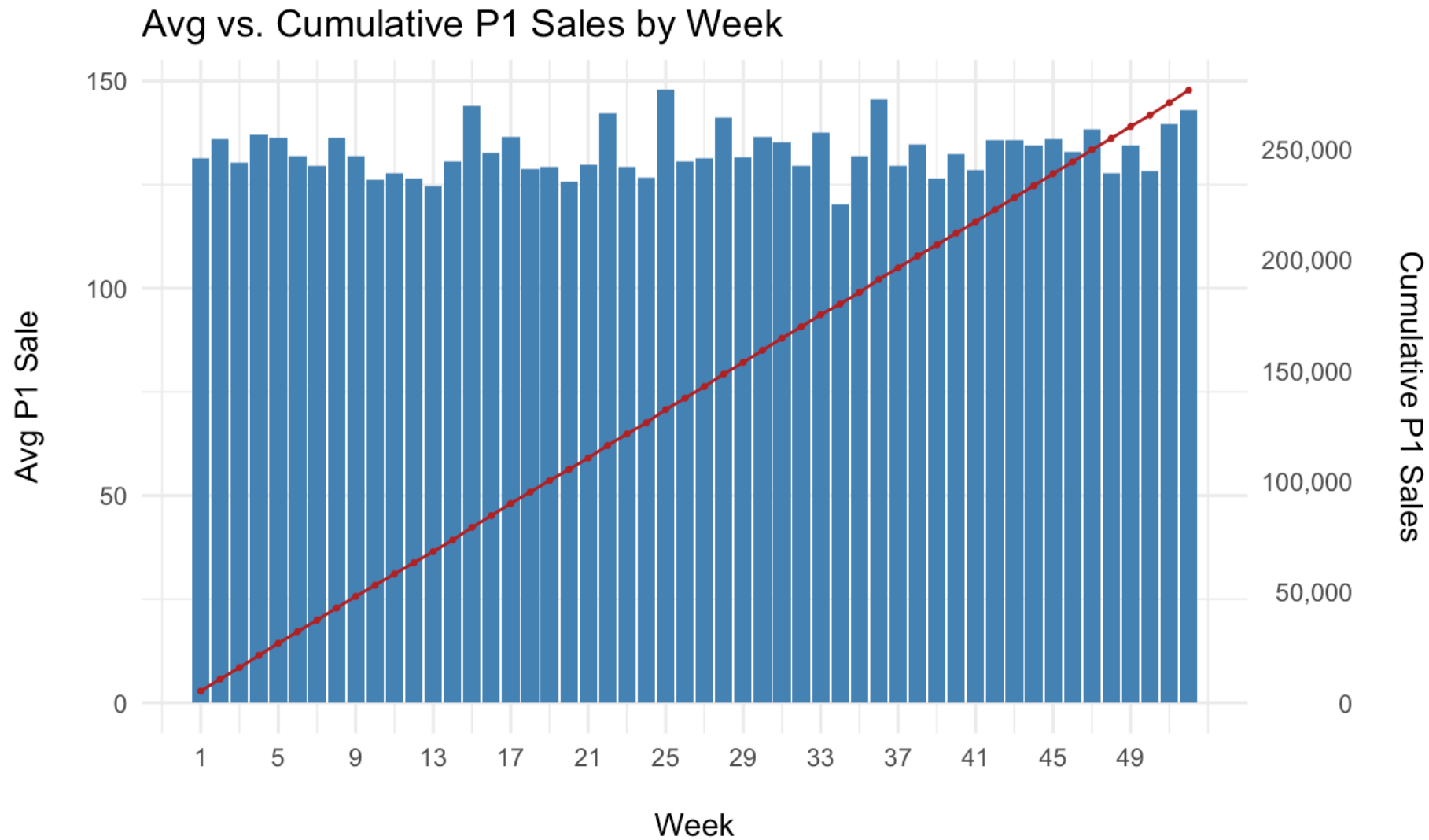


Area chart



Stacked areas show composition over time, but only the bottom series sits on a flat baseline, so read the others with care.

Bar + line combo



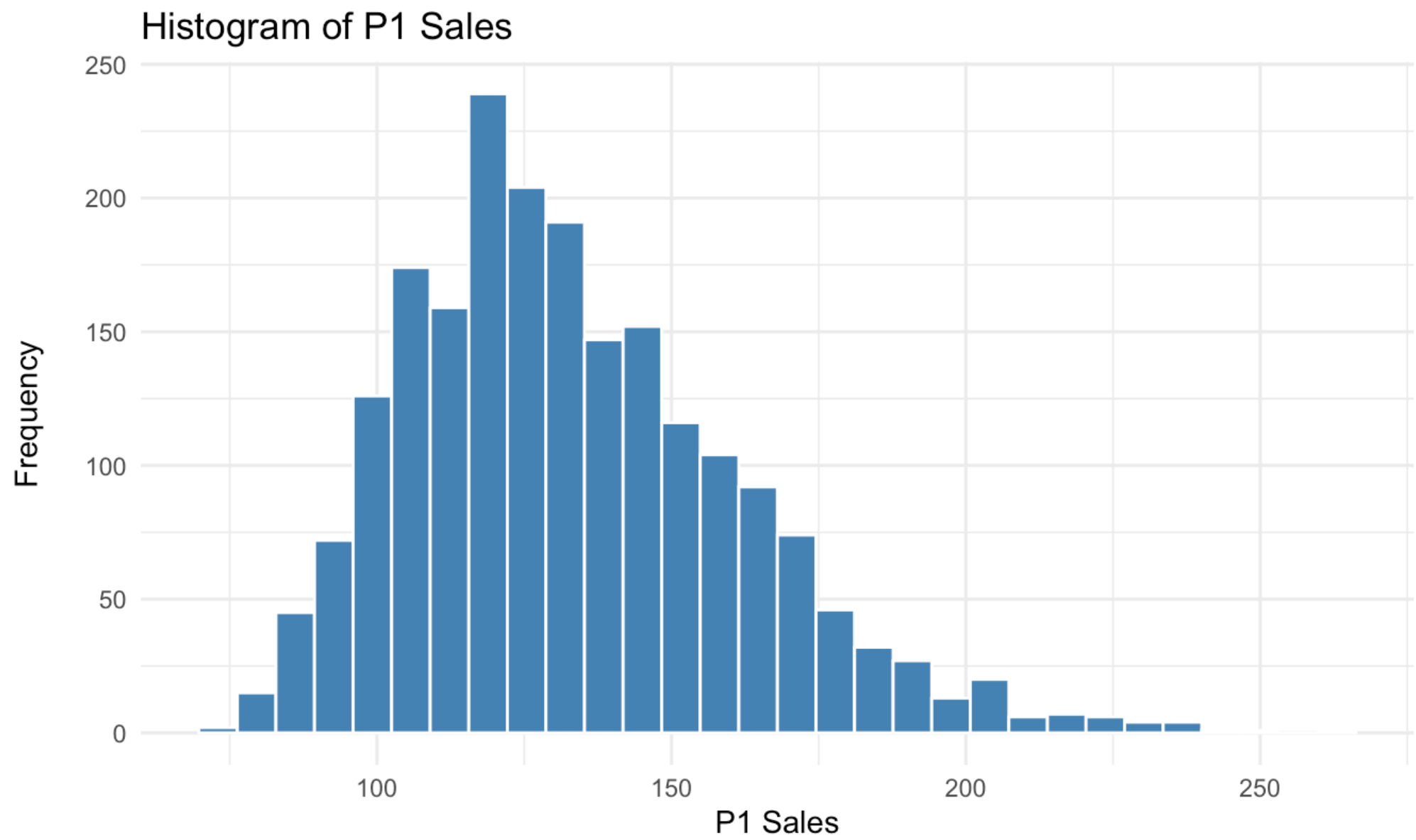
Two scales on one chart: level (bars, left axis) vs. running total (line, right axis).

Distribution & density

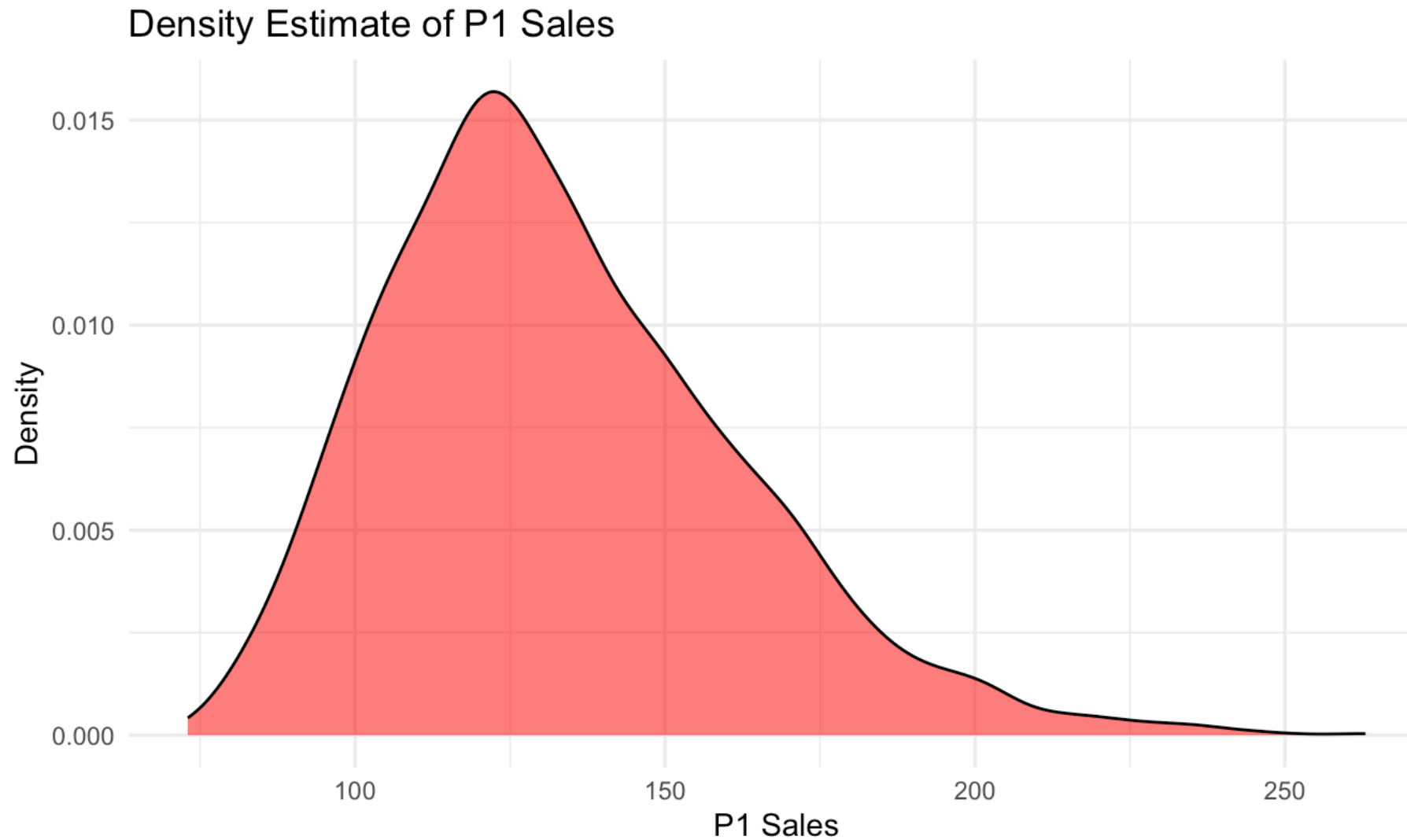
Understand the shape, spread, and outliers of a variable.

- Histogram
- Density plot
- Box plot
- Violin plot

Histogram

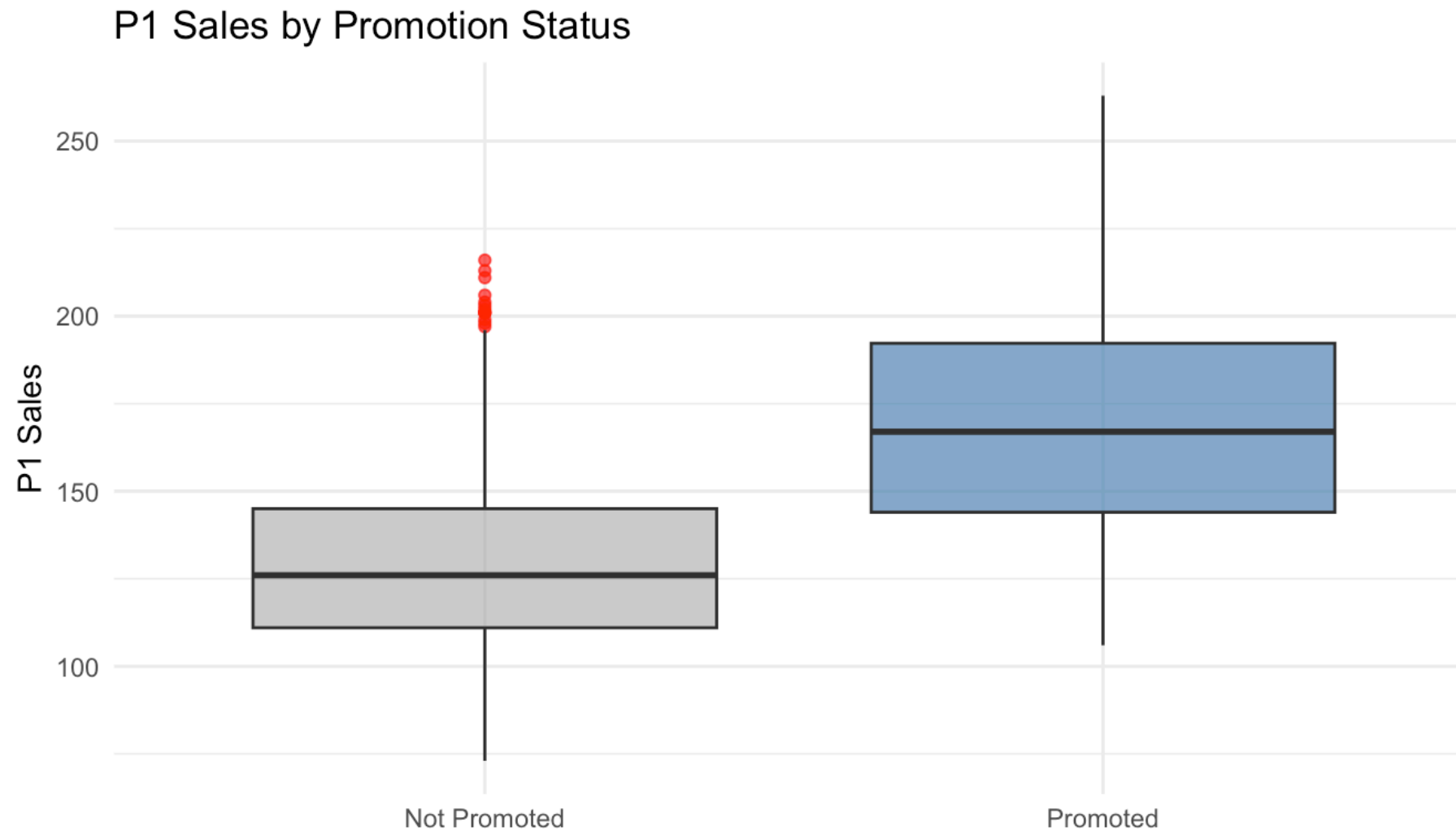


Density plot



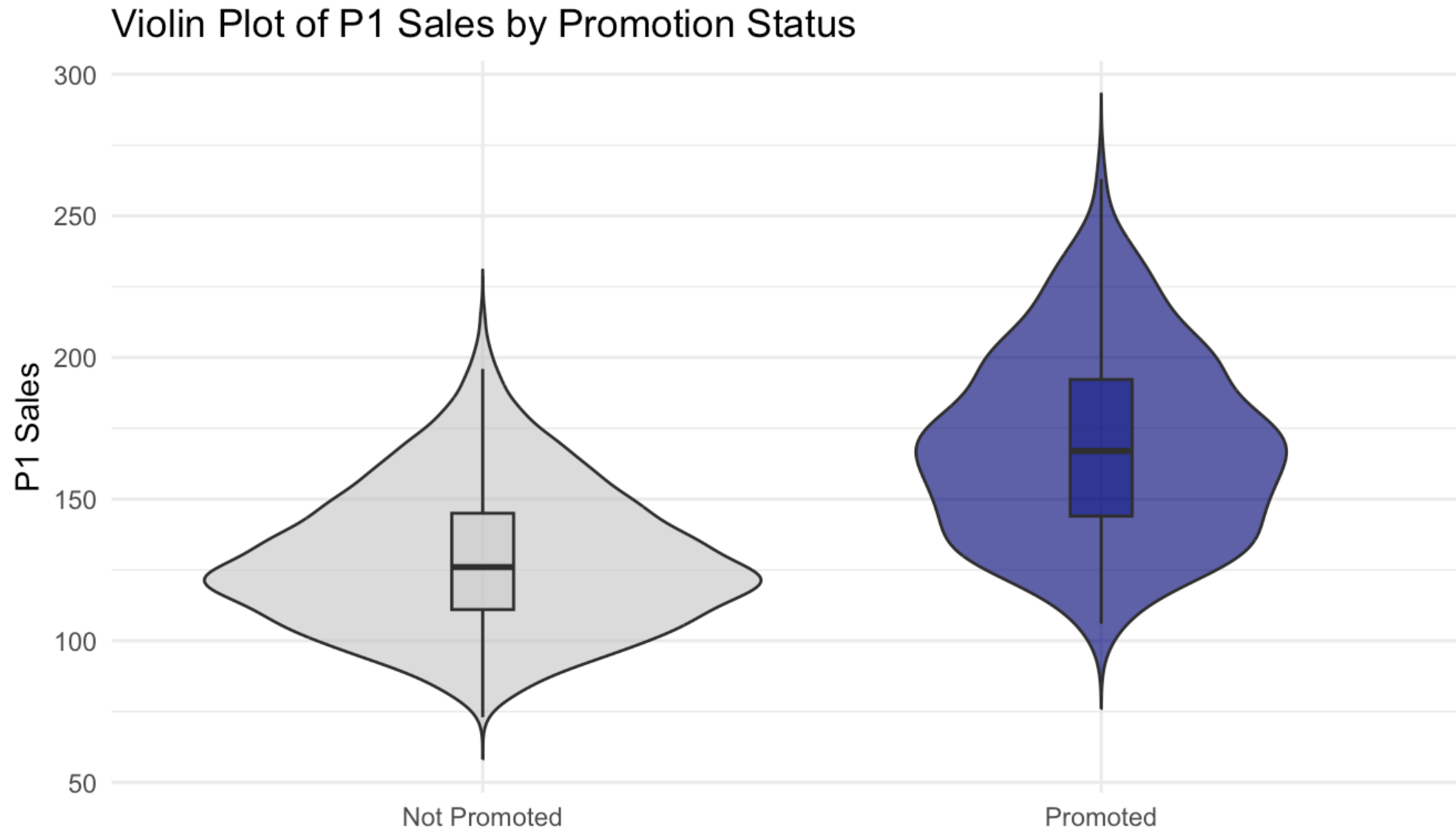
A smoothed histogram: highlights modes, hides bin-choice artifacts.

Box plot



Median, quartiles, whiskers, outliers: a five-number summary per group.

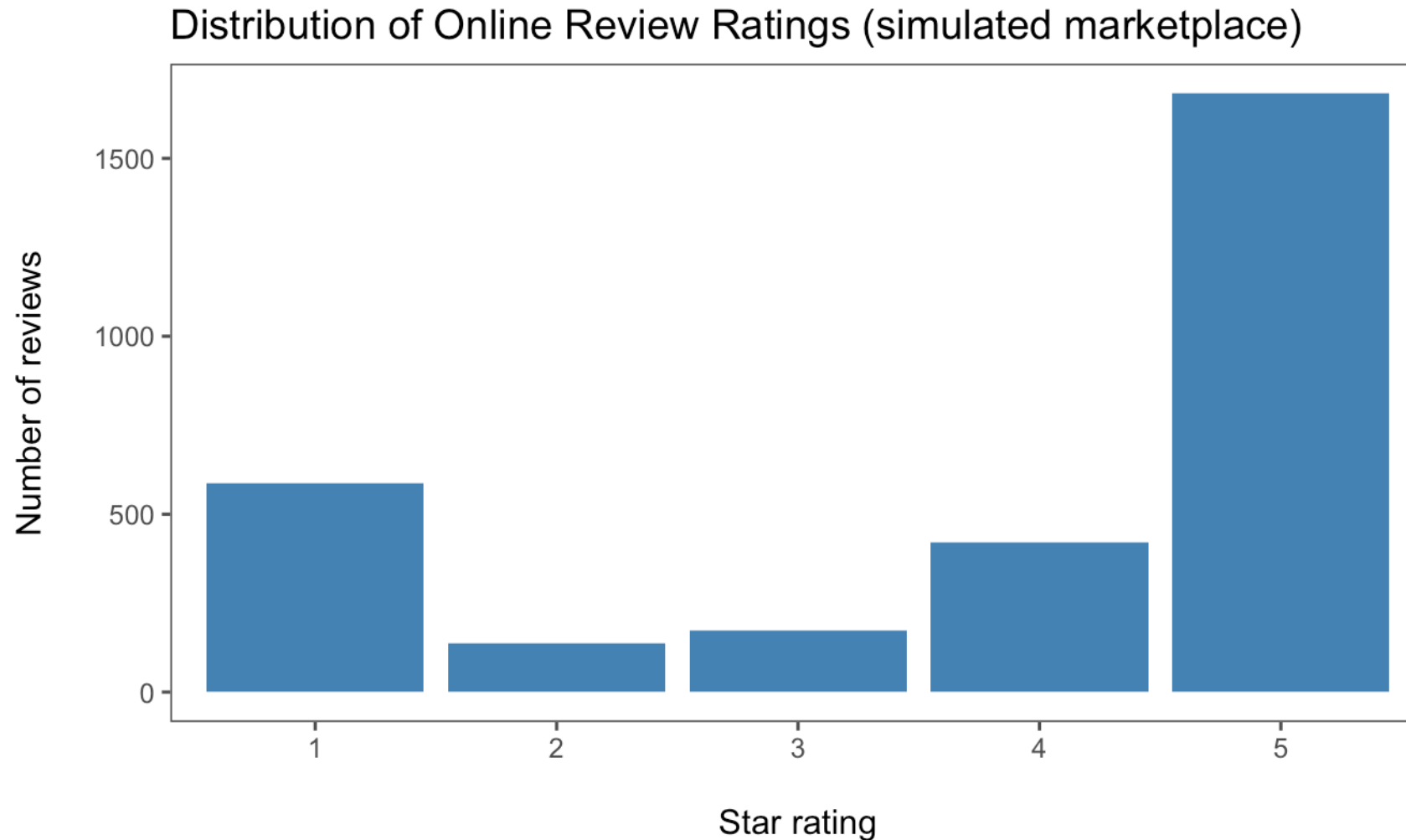
Violin plot



Box plot vs violin plot

Aspect	Boxplot	Violin plot
What it shows	Five-number summary (Q1, median, Q3; whiskers; outliers)	Distribution shape (smoothed density)
Outliers	Explicit points beyond whiskers (1.5×IQR rule)	Not shown by default
Multimodality	Hard to see	Easy to see (multiple “bulges”)
Robustness	Robust: based on quantiles	Depends on bandwidth and smoothing
Small samples	Reliable	Can be misleading (noisy density)
When to use	Compare medians/spread cleanly	Understand shape beyond the median

Distributions in the wild: online ratings



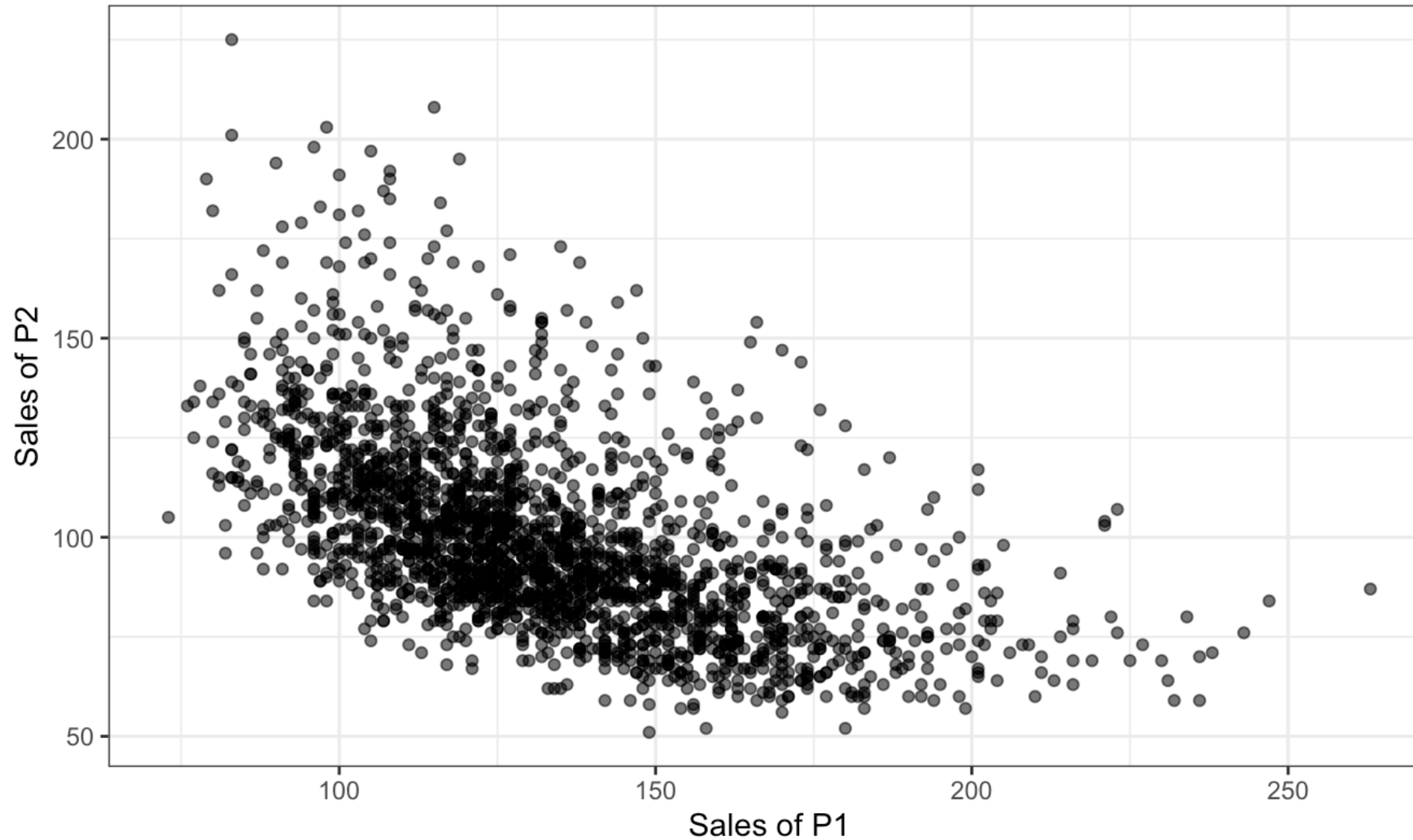
Review ratings are typically **J-shaped**: delighted and furious customers review, the indifferent middle stays silent. The mean rating hides this: plot the distribution. Much more on reviews and reputation later in the course.

Relationships & correlation

Explore how two (or more) variables move together.

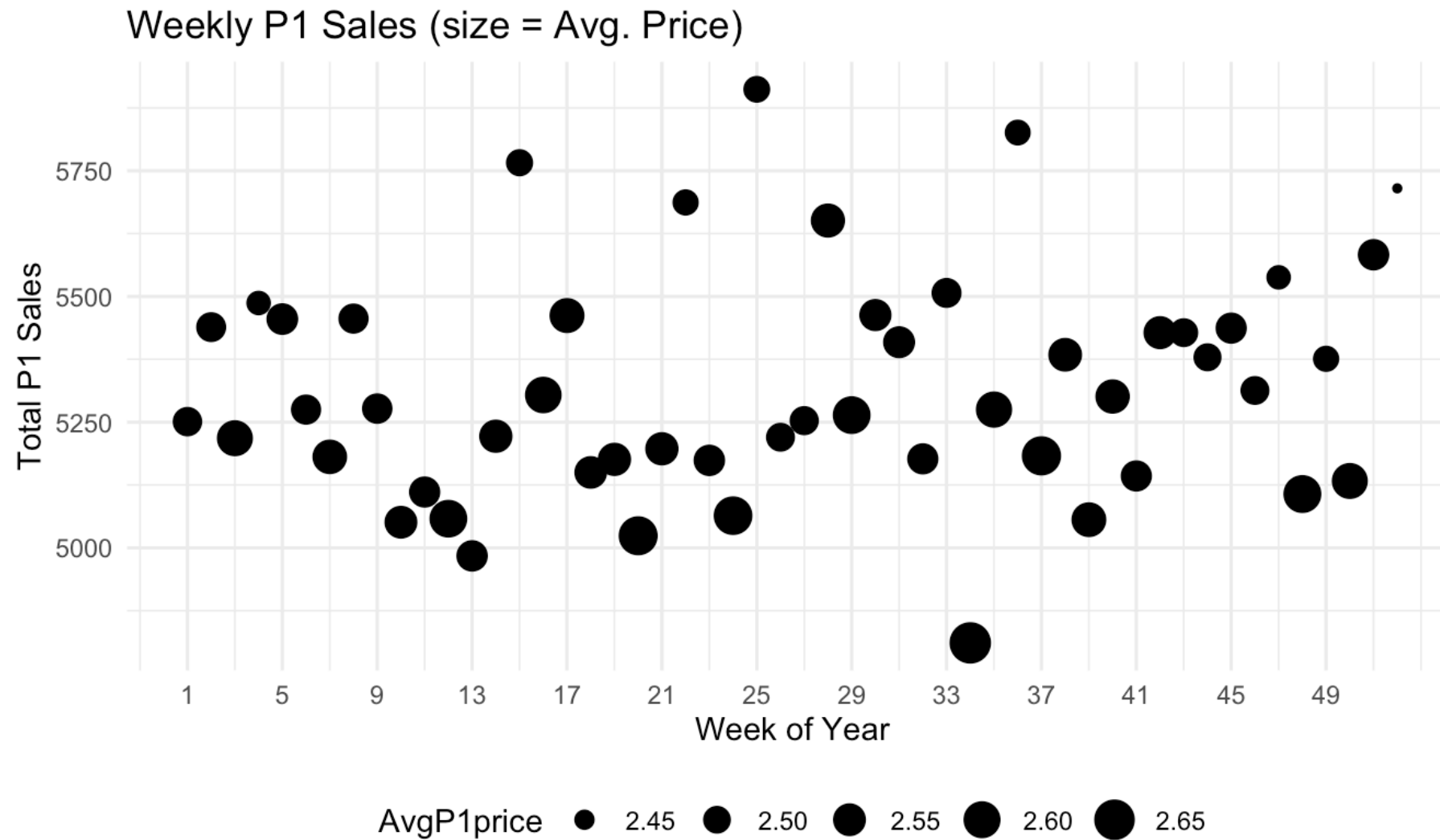
- Scatter plot
- Bubble chart (scatter + size)

Scatter plot



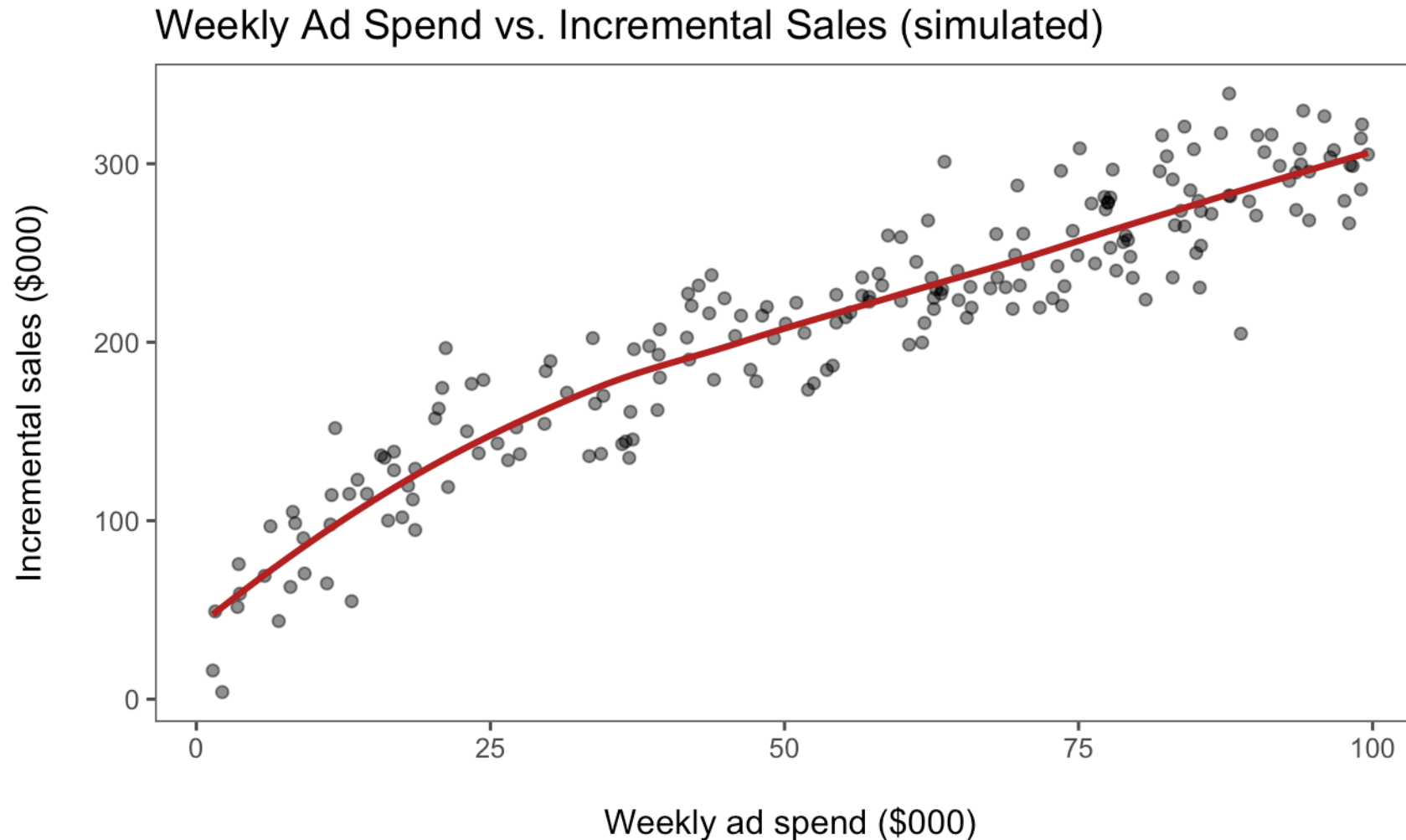
P1 and P2 sales are negatively associated. What marketing story could explain that?

Bubble chart



A third variable mapped to point size: do high-sales weeks coincide with low prices?

Ad spend and diminishing returns



The response curve is **concave**: doubling spend does not double sales. A scatter plus a smoother is often the first diagnostic for budget-allocation questions.

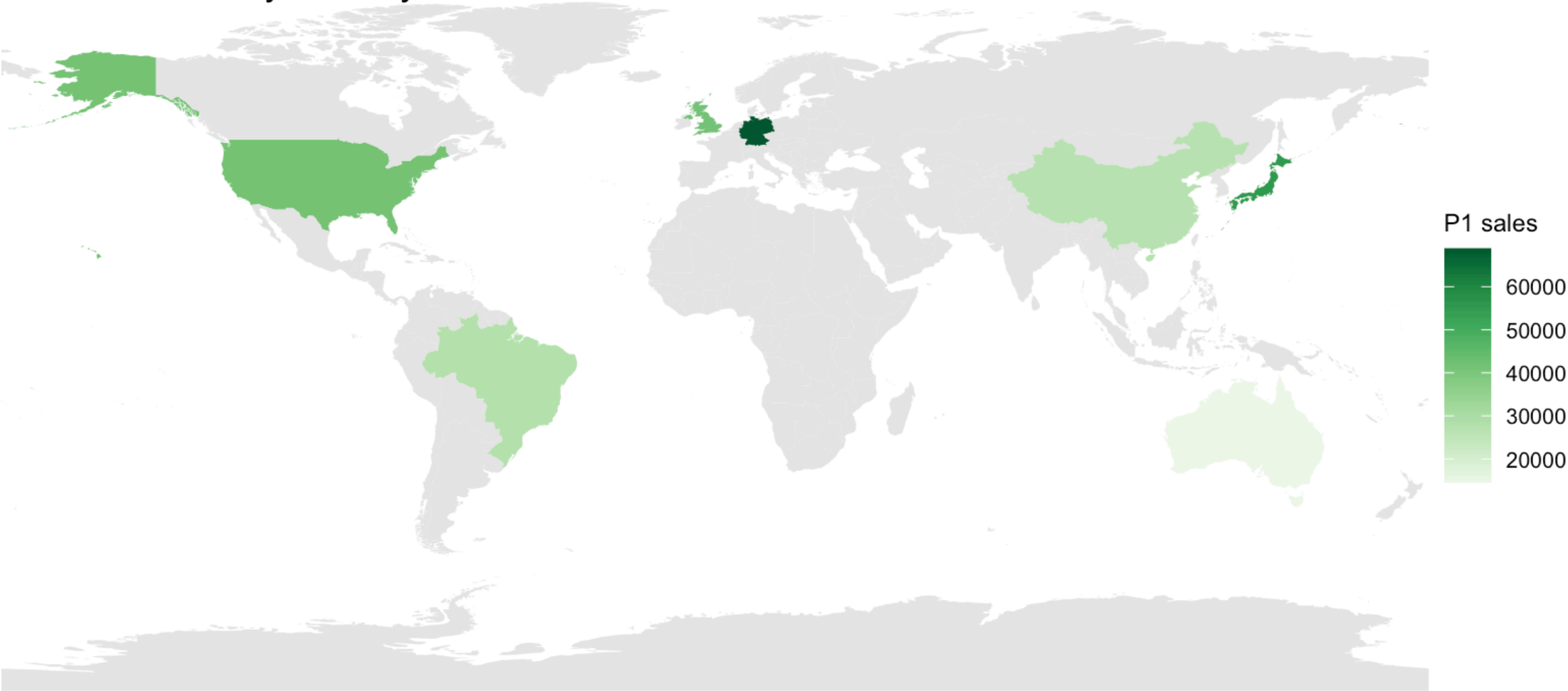
Geospatial & matrix data

Map values over space or grid layouts.

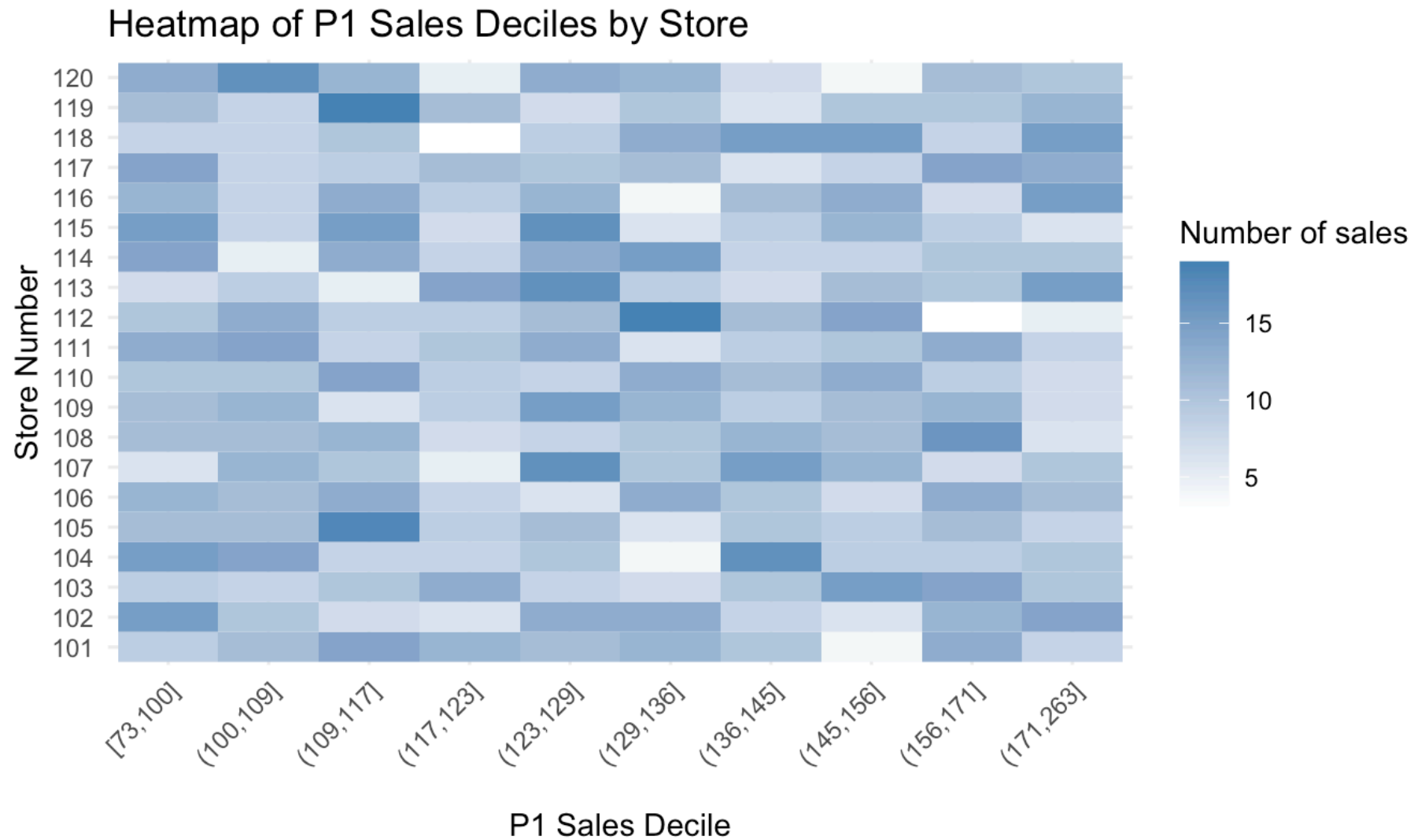
- Geospatial map (choropleth, points)
- Heatmap (matrix of values)

Choropleth map

Total P1 Sales by Country

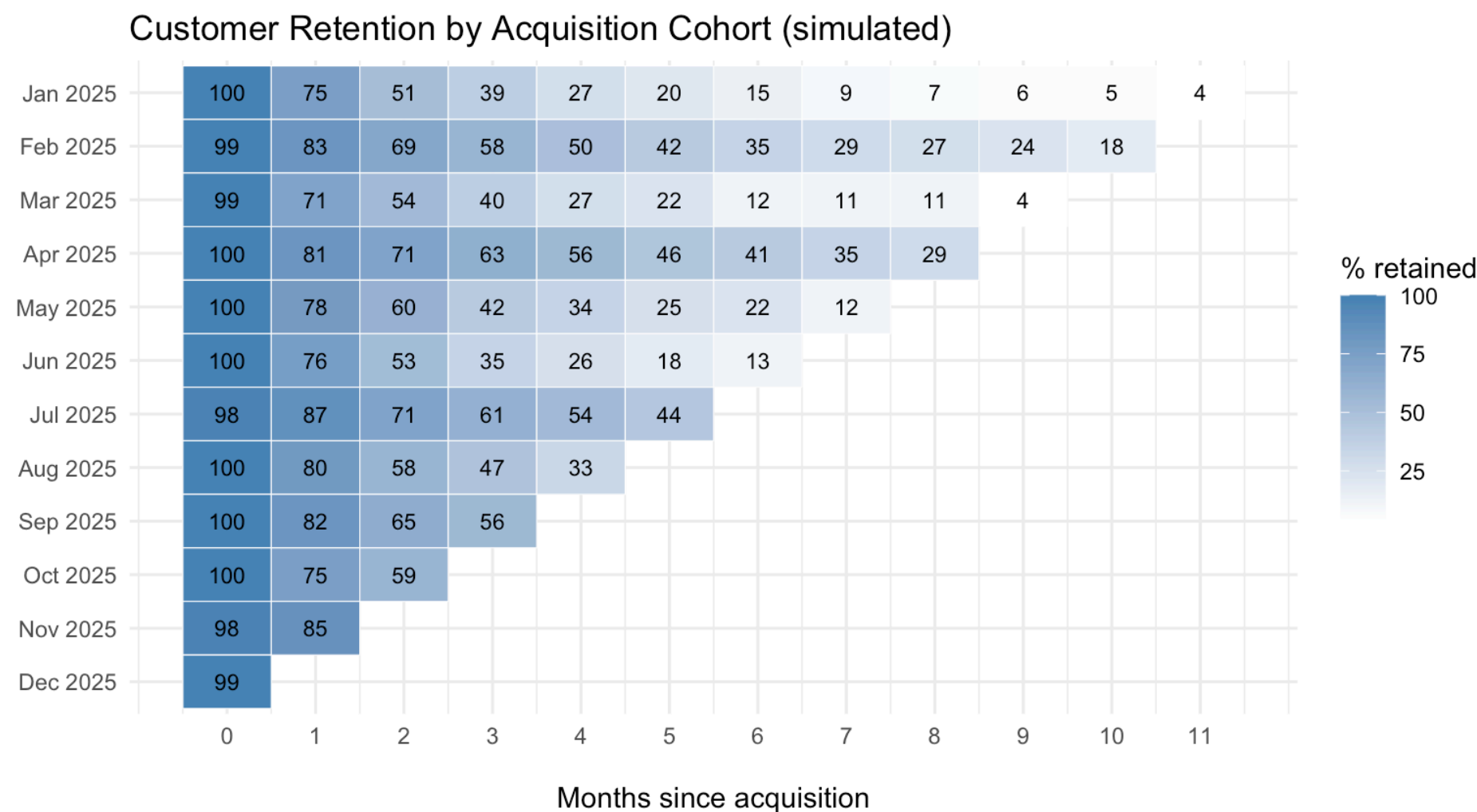


Heatmap



Two categorical axes + a color-encoded value: spot which stores skew high or low.

The marketing classic: cohort retention



Rows are acquisition cohorts, columns are months since signup: the standard view for subscription businesses. Are newer cohorts retaining better or worse?

Choosing the Best Chart

Choosing the best chart

1. **Define your question:** Comparison? Trend? Distribution? Relationship?
2. **Inspect your variables:** categorical vs. numeric; panel vs. cross-section; time vs. location vs. hierarchy

For example:

- Price **trends over time** → line chart
- Compare current **job-post counts across languages** → bar chart
- Explore **salary distributions by city** → box or violin plot

Best Practices

1. Simplify and declutter

- Reduce “chart junk”: eliminate unnecessary gridlines, backgrounds, and 3D effects
 - I often use `theme_few()` in R
- Legends only when needed: if you label directly on the plot, drop the legend; don't be redundant

2. Readable scales and labels

- **Descriptive titles and subtitles:** tell viewers what they're looking at and why it matters
- **Clear axis labels:** include units (e.g., "Sales (USD Millions)") and avoid abbreviations when possible
- **Consistent breaks:** choose nice, round numbers or evenly spaced dates
- Always add **figure notes** at the bottom of the figure in documents and reports

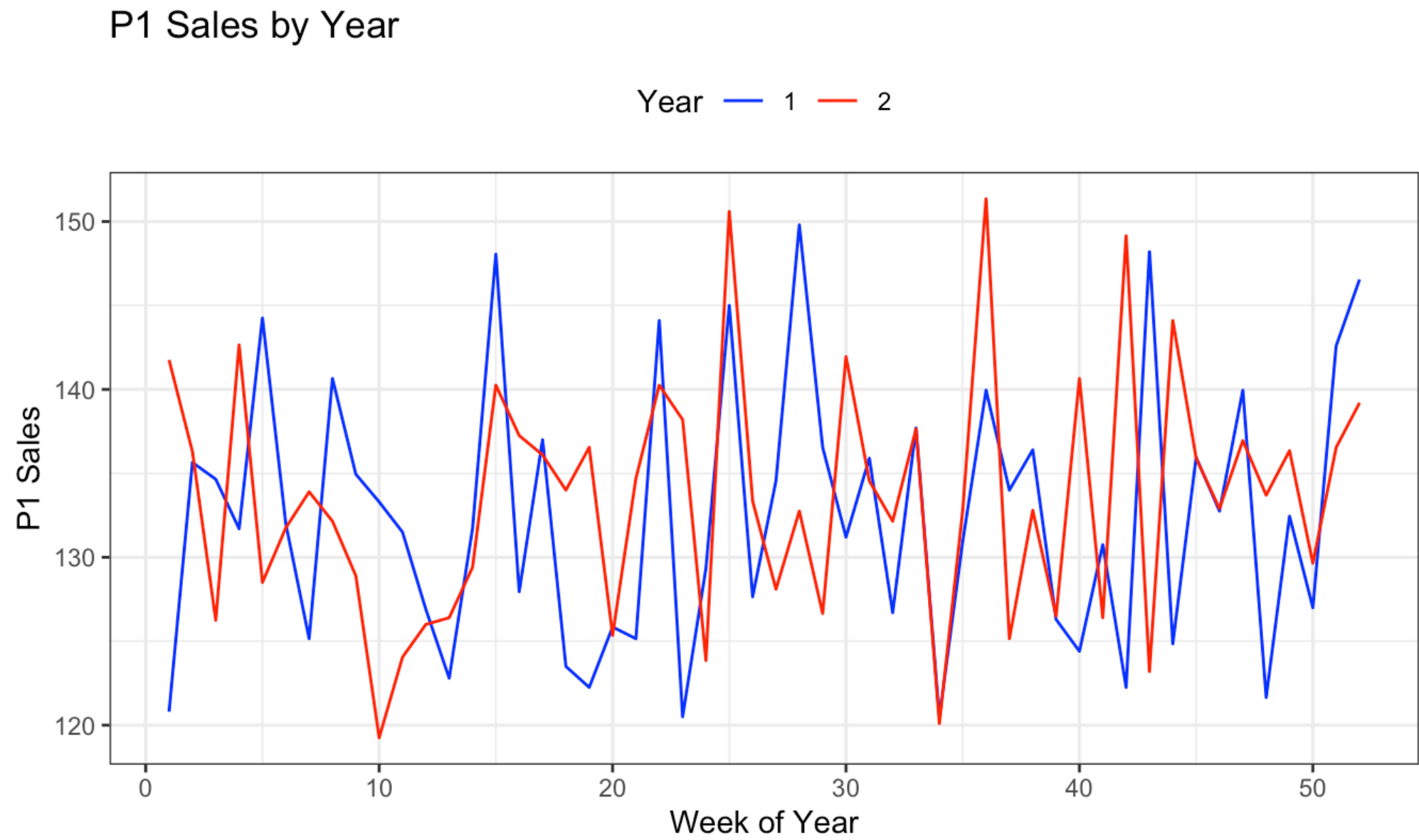
3. Accessible color and style

- **Color-blind-friendly palettes:** in R, `RColorBrewer` (“Set2”, “Dark2”) or `viridis`
- **Limit colors:** no more than 4–6 distinct colors in a single plot; for many categories, consider facets or small multiples instead
- **Transparency** to manage overplotting in dense scatter or area charts (the `alpha` parameter in R)

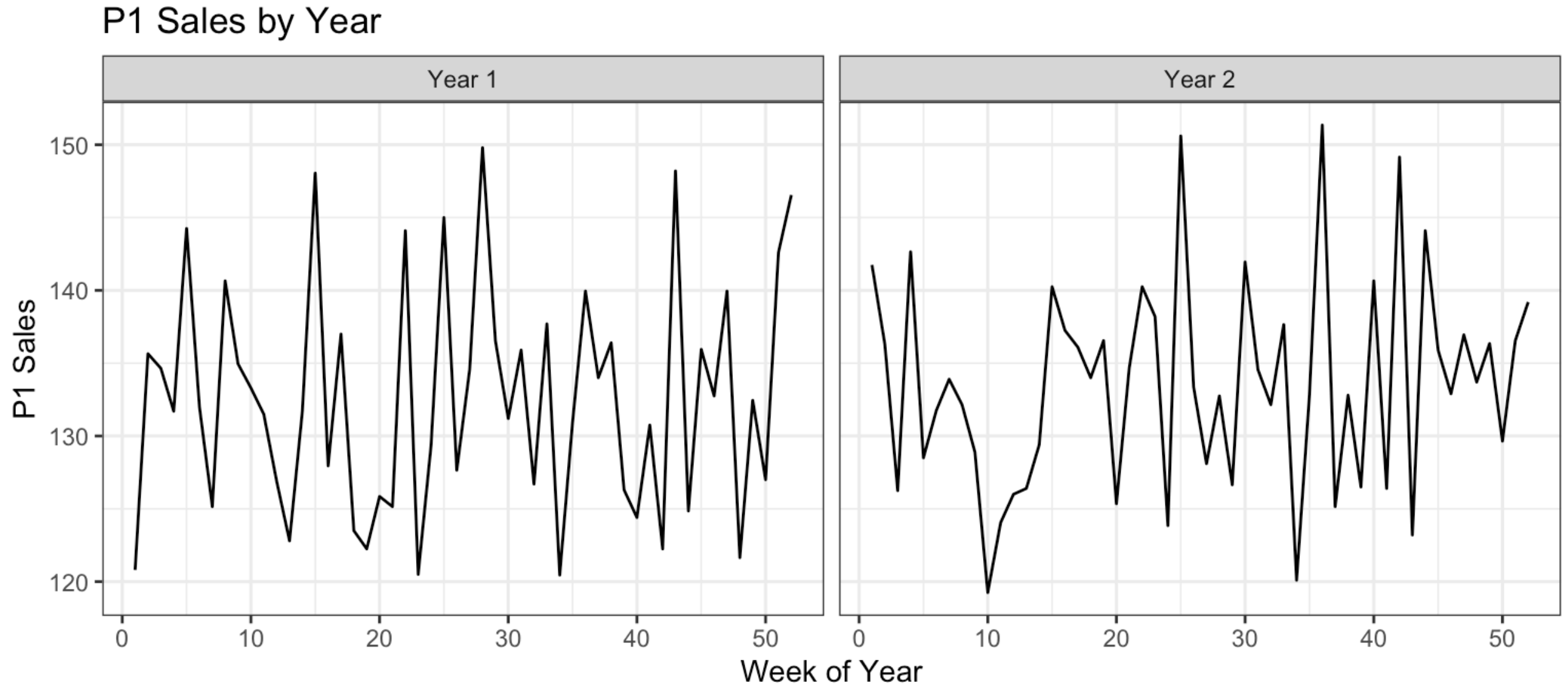
4. Leverage facets

- `facet_wrap()` / `facet_grid()` split the data by a categorical variable rather than cramming everything into one panel
- Ensures consistent scales and easy side-by-side comparisons

Grouping: everything in one panel

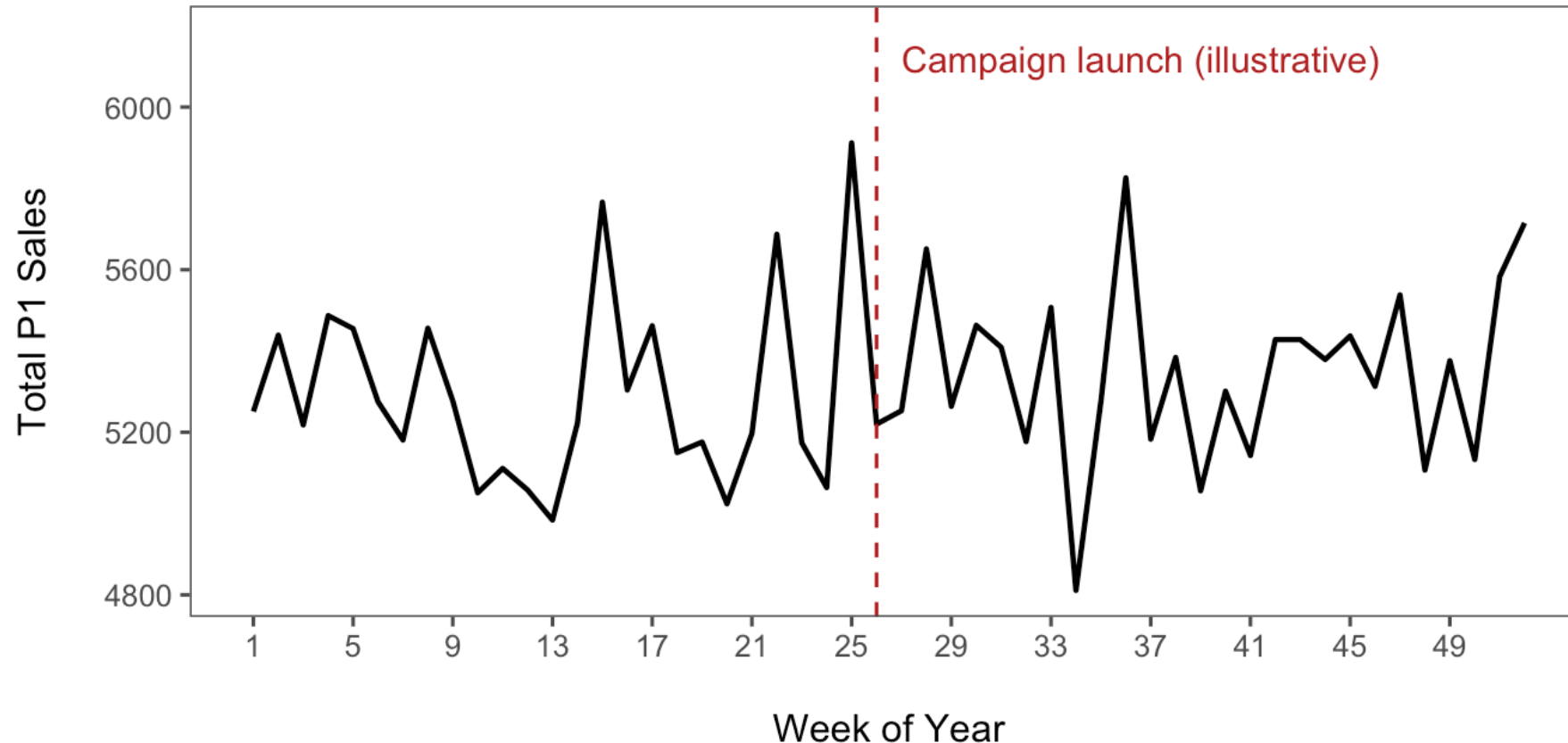


Faceting: one panel per group



5. Annotate and highlight key insights

- **Direct labels** with `geom_text()` or `ggrepel` for calling out peaks, thresholds, or outliers
- **Annotations** (`annotate()`, `geom_vline()`/`geom_hline()`) to mark events: product launches, policy changes, seasonal holidays



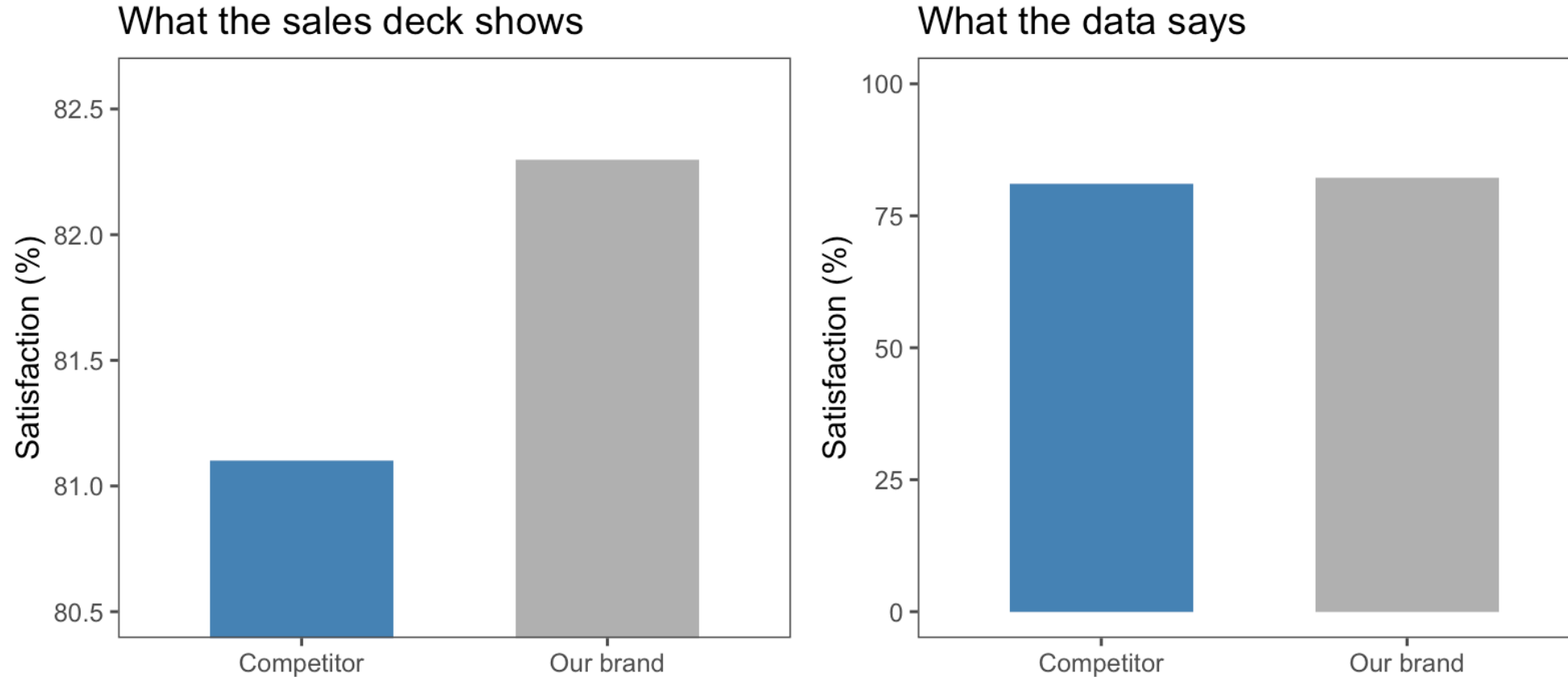
6. Consistency across plots

- Define a **custom theme** and apply it to every chart so colors, fonts, and margins are consistent
- Use the **same color mapping** for the same variables across multiple plots

7. Validate and iterate

- **Peer review:** show rough drafts to classmates. Do they “get” the story without explanation?
- **Test in grayscale:** verify that patterns and contrasts remain readable when printed without color

How to mislead with a chart



Same data, different y-axis. Truncated axes exaggerate differences, a favorite trick of dashboards and ad claims. When you see a bar chart, **check where the axis starts.**

Rules are made to be broken

Break the rules only when doing so tells a **clearer story**. Good visualization is as much art as science!

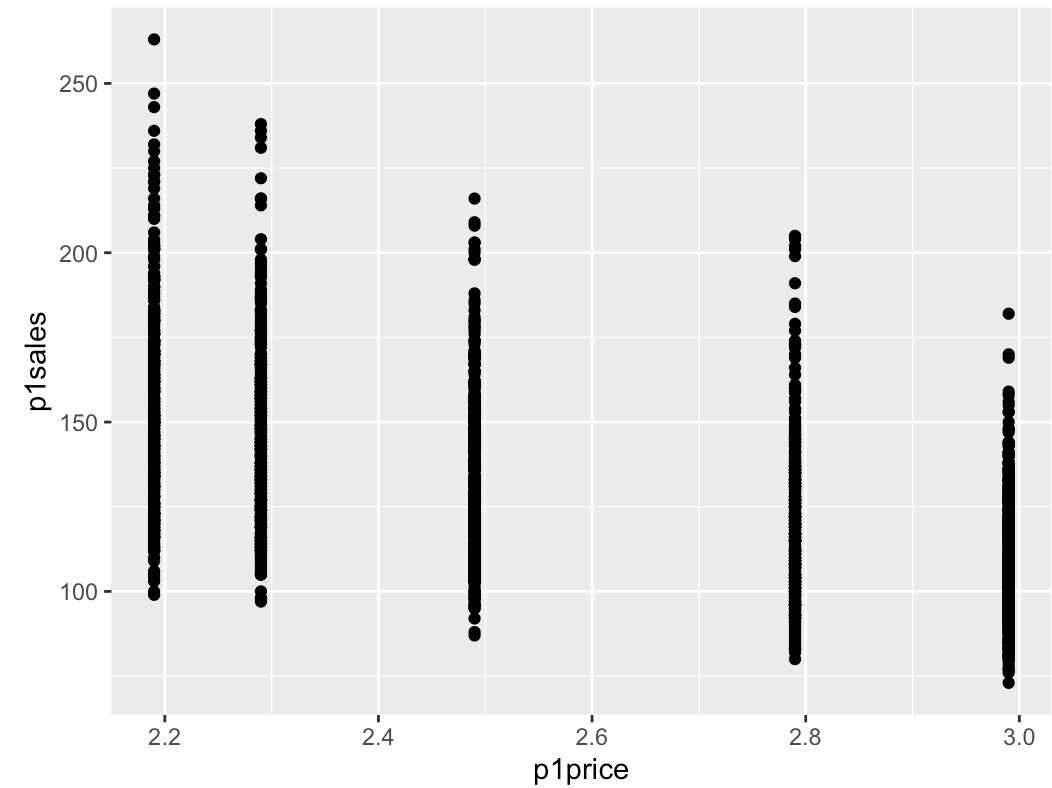
Hands-On: Beautifying a Figure

Our example: price and sales

- A scatterplot is the natural tool to understand the **relationship between two variables**
- Let's use it on the most fundamental relationship in marketing, the **demand curve**: price (`p1price`) vs. weekly unit sales (`p1sales`) of product 1 in the store data

Step 0: the default plot

```
1 ggplot(data = store.df) +  
2   geom_point(  
3     mapping = aes(x = p1price, y = p1sales)  
4   )
```



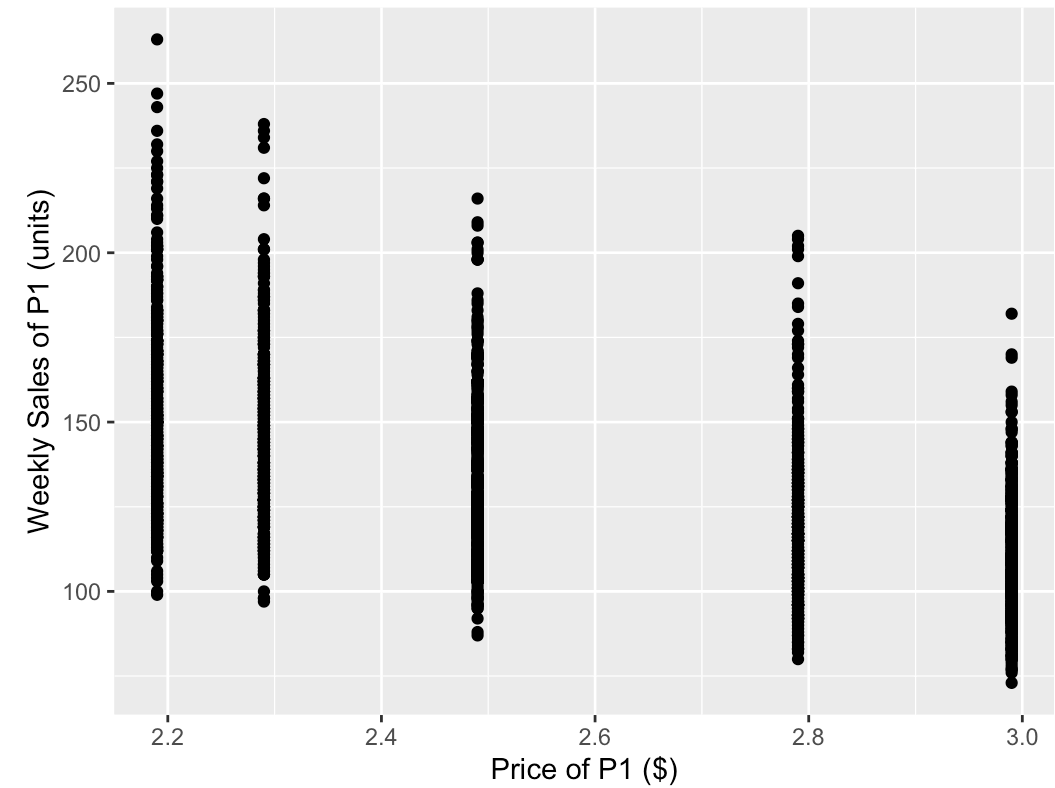
What is wrong with it?

- Axis labels are **not easy to interpret** (what is `p1price`?)
- The **units** in which the variables are measured are unclear
- **Overplotting**: prices take a few discrete values, so points pile up on top of each other and hide the density
- Axis **font is very small**, with little space between labels and axis names

Let's fix these issues...

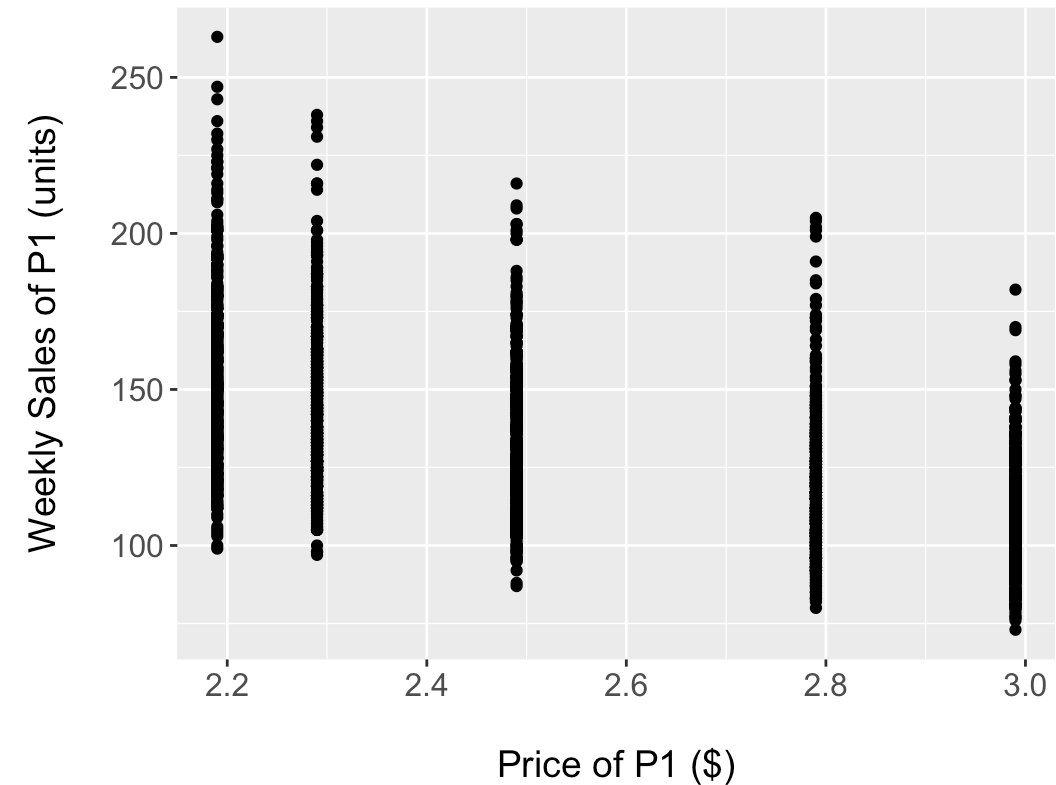
Step 1: label the axes

```
1 ggplot(data = store.df) +  
2   geom_point(  
3     mapping = aes(x = p1price, y = p1sales)  
4   ) +  
5   labs(  
6     x = "Price of P1 ($)",  
7     y = "Weekly Sales of P1 (units)"  
8   )
```



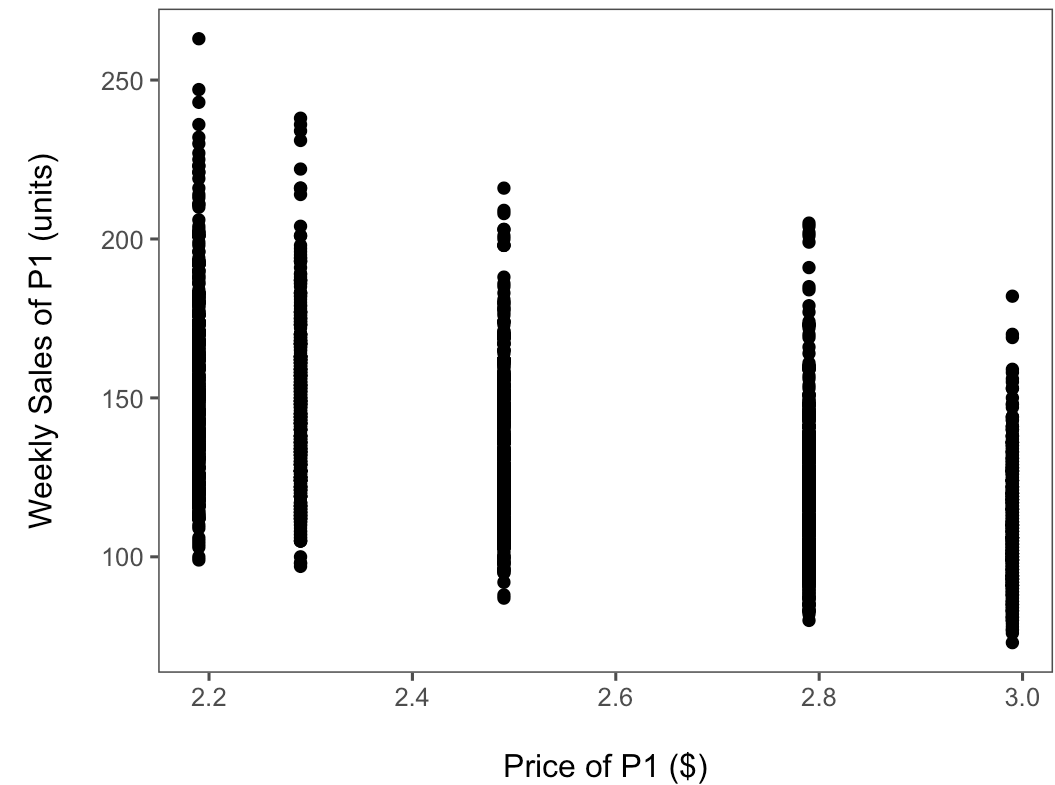
Step 2: larger fonts, more breathing room

```
1 ggplot(data = store.df) +  
2   geom_point(  
3     mapping = aes(x = p1price, y = p1sales)  
4   ) +  
5   labs(  
6     x = "\nPrice of P1 ($)",  
7     y = "Weekly Sales of P1 (units)\n"  
8   ) +  
9   theme(  
10    axis.title.x = element_text(size = 14),  
11    axis.title.y = element_text(size = 14),  
12    axis.text.x  = element_text(size = 12),  
13    axis.text.y  = element_text(size = 12)  
14  )
```



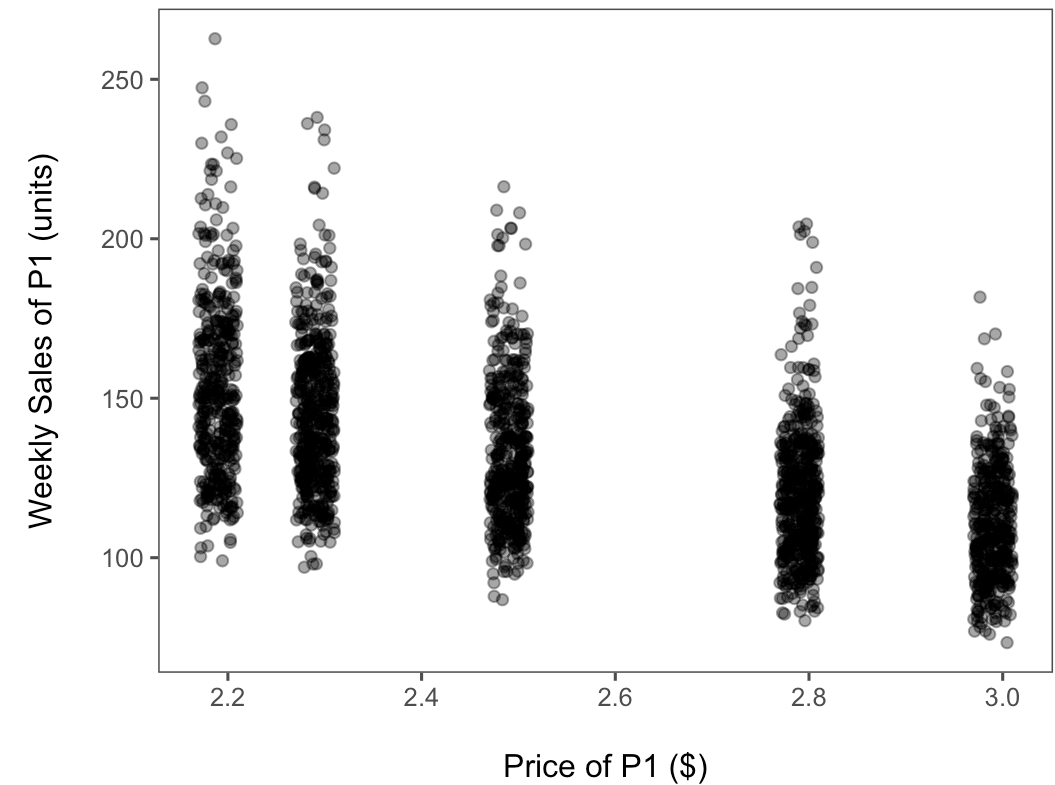
Step 3: a cleaner theme

```
1 ggplot(data = store.df) +  
2   geom_point(  
3     mapping = aes(x = p1price, y = p1sales)  
4   ) +  
5   labs(  
6     x = "\nPrice of P1 ($)",  
7     y = "Weekly Sales of P1 (units)\n"  
8   ) +  
9   theme(  
10    axis.title.x = element_text(size = 14),  
11    axis.title.y = element_text(size = 14),  
12    axis.text.x  = element_text(size = 12),  
13    axis.text.y  = element_text(size = 12)  
14  ) +  
15  theme_few()
```



Step 4: fix the overplotting

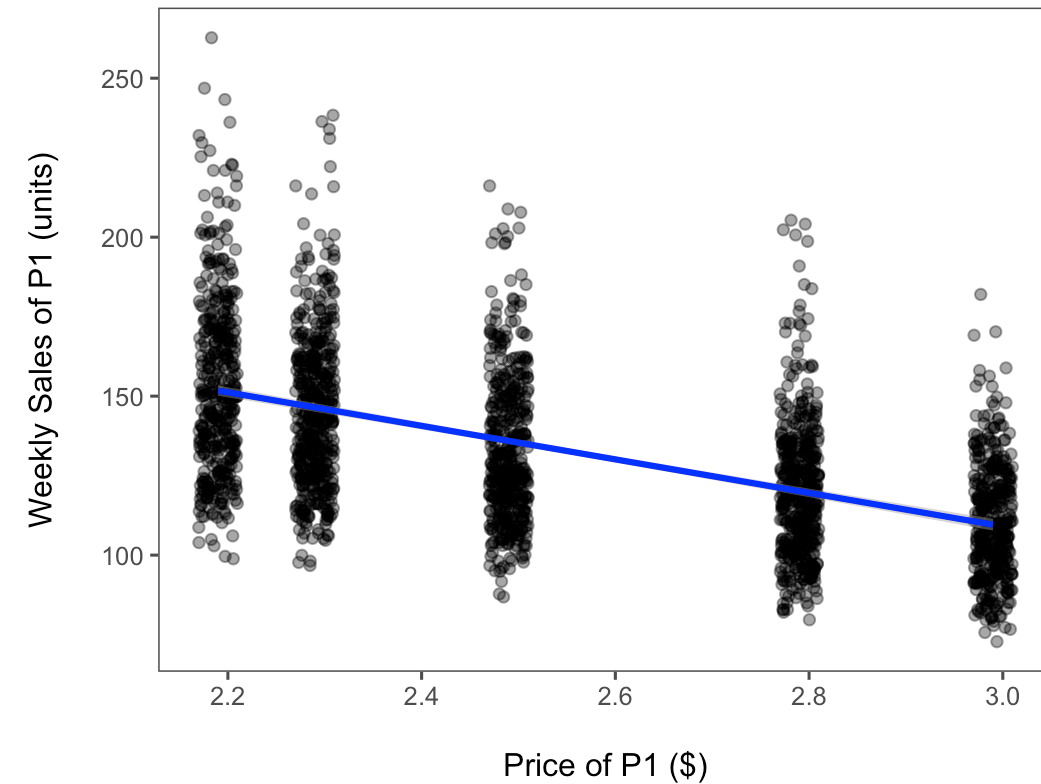
```
1 ggplot(data = store.df) +  
2   geom_jitter(  
3     mapping = aes(x = p1price, y = p1sales),  
4     width = 0.02, alpha = 0.4  
5   ) +  
6   labs(  
7     x = "\nPrice of P1 ($)",  
8     y = "Weekly Sales of P1 (units)\n"  
9   ) +  
10  theme(  
11    axis.title.x = element_text(size = 14),  
12    axis.title.y = element_text(size = 14),  
13    axis.text.x = element_text(size = 12),  
14    axis.text.y = element_text(size = 12)  
15  ) +  
16  theme_few()
```



`geom_jitter()` adds a little horizontal noise; `alpha` makes overlaps visible.

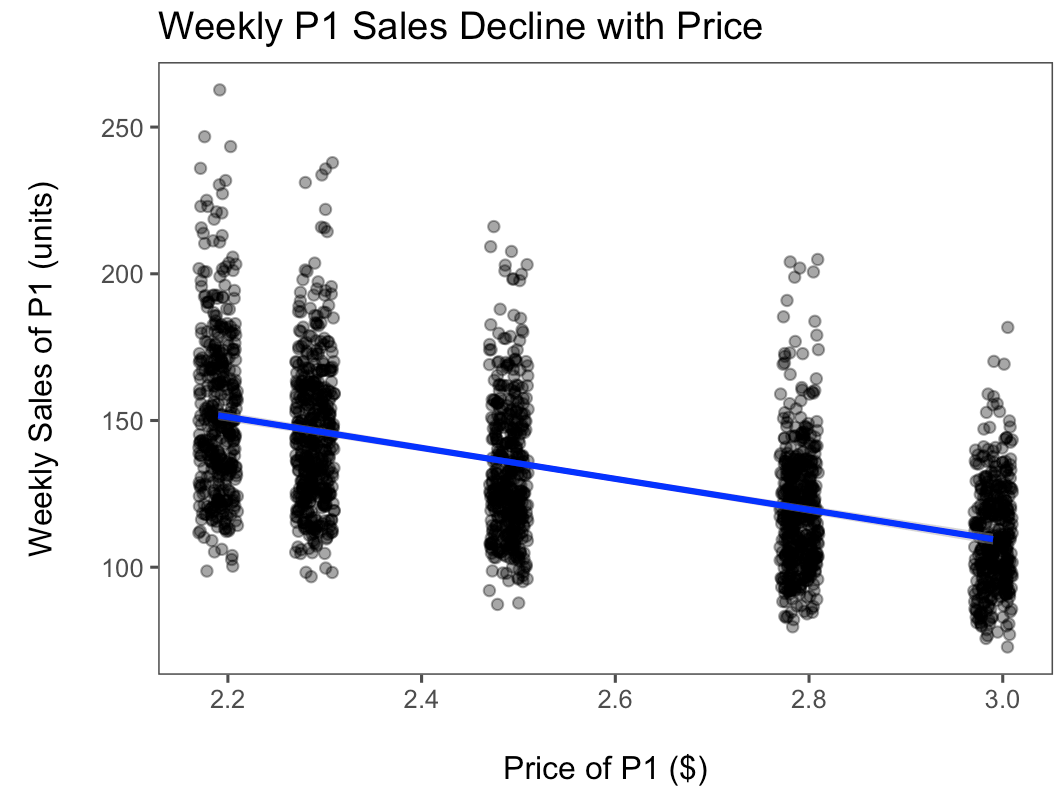
Step 5: add the trend

```
1 ggplot(data = store.df) +  
2   geom_jitter(  
3     mapping = aes(x = p1price, y = p1sales),  
4     width = 0.02, alpha = 0.4  
5   ) +  
6   labs(  
7     x = "\nPrice of P1 ($)",  
8     y = "Weekly Sales of P1 (units)\n"  
9   ) +  
10  theme(  
11    axis.title.x = element_text(size = 14),  
12    axis.title.y = element_text(size = 14),  
13    axis.text.x = element_text(size = 12),  
14    axis.text.y = element_text(size = 12)  
15  ) +  
16  geom_smooth(  
17    mapping = aes(x = p1price, y = p1sales),  
18    method = "lm", color = "blue"  
19  ) +
```



Step 6: add a title

```
1 ggplot(data = store.df) +  
2   geom_jitter(  
3     mapping = aes(x = p1price, y = p1sales),  
4     width = 0.02, alpha = 0.4  
5   ) +  
6   labs(  
7     x = "\nPrice of P1 ($)",  
8     y = "Weekly Sales of P1 (units)\n",  
9     title = "Weekly P1 Sales Decline with Price"  
10  ) +  
11  theme(  
12    axis.title.x = element_text(size = 14),  
13    axis.title.y = element_text(size = 14),  
14    axis.text.x = element_text(size = 12),  
15    axis.text.y = element_text(size = 12)  
16  ) +  
17  geom_smooth(  
18    mapping = aes(x = p1price, y = p1sales),  
19    method = "lm", color = "blue"
```



A downward-sloping demand curve. How much sales fall when price rises is the **price elasticity**; we will estimate it with regression in week 4.

Wrap-up

Code and data to reproduce today's charts, all on the course website:

- [w1-code.zip](#): one-click download of everything below (scripts + data)
- [w1-2-chart-types-class.R](#): every chart type shown today
- [w1-2-data-viz-beautify.R](#): the beautification walkthrough
- [w1-2-simulate-datasets.R](#): generates the simulated datasets
- [data/](#): [store-sales.csv](#) (real store data) and five small simulated CSVs: customer revenue, online ratings, ad spend, cohort retention, job postings

Questions?